

Stream cipher and Block Cipher

Chapter-3: Block Ciphers and the Data Encryption Standard

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1. Stream ciphers and block ciphers
2. Block Cipher Principles
3. Data Stream ciphers and block ciphers
4. Confusion & Diffusion
5. Data Encryption Standard (DES)
6. Avalanche Effect
7. Strength of DES
8. Design principles of block cipher

Stream cipher and Block Cipher:

A stream cipher is one that encrypts a digital data stream one bit or one byte at a time.

Examples:

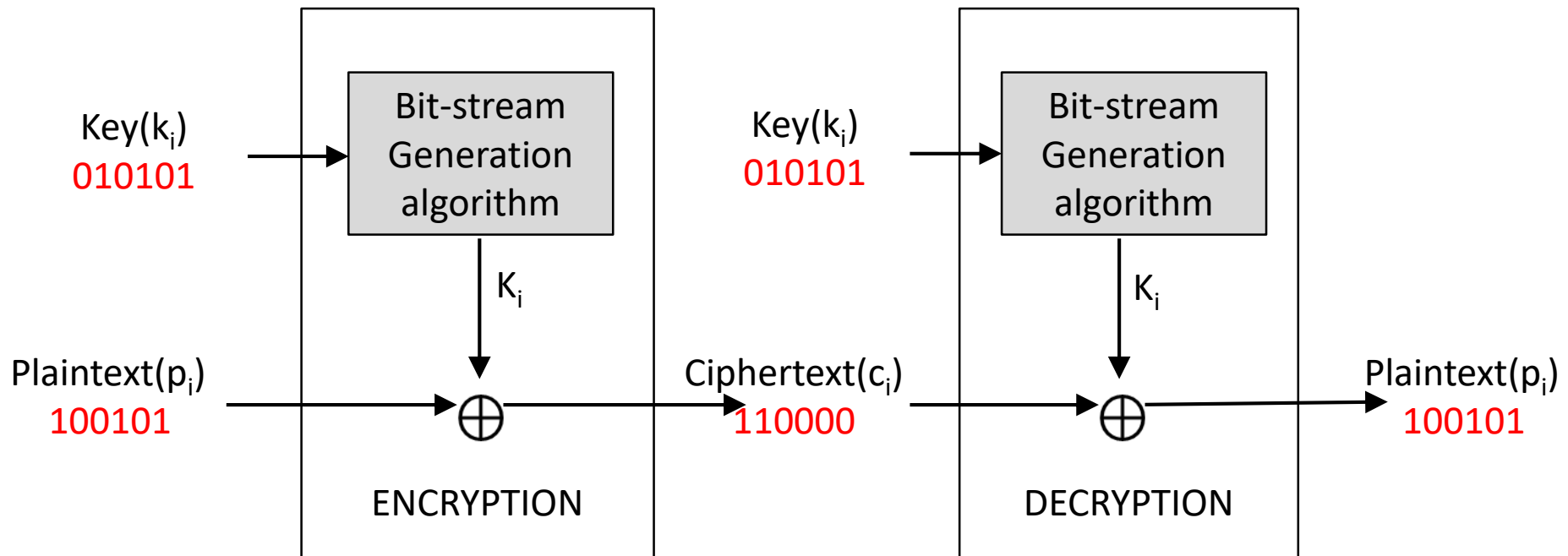
Autokeyed Vigenère cipher

A5/1

RC4

Vernam cipher.

Stream cipher and Block Cipher:



Stream cipher and Block Cipher:

A block cipher is one in which a block of plaintext is treated as a whole and used to produce a ciphertext block of equal length. Typically, a block size of 64 or 128 bits is used.

Examples:

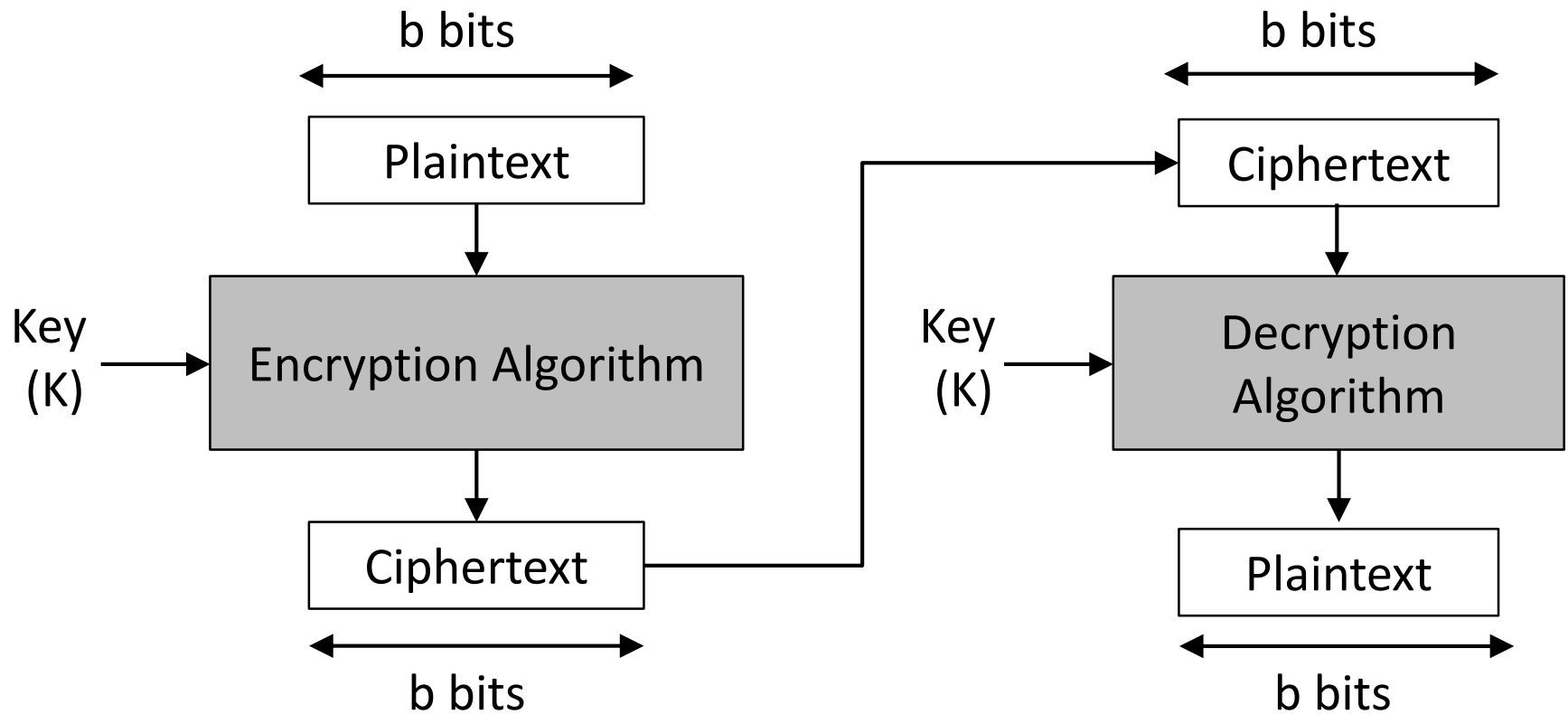
Feistel cipher

DES

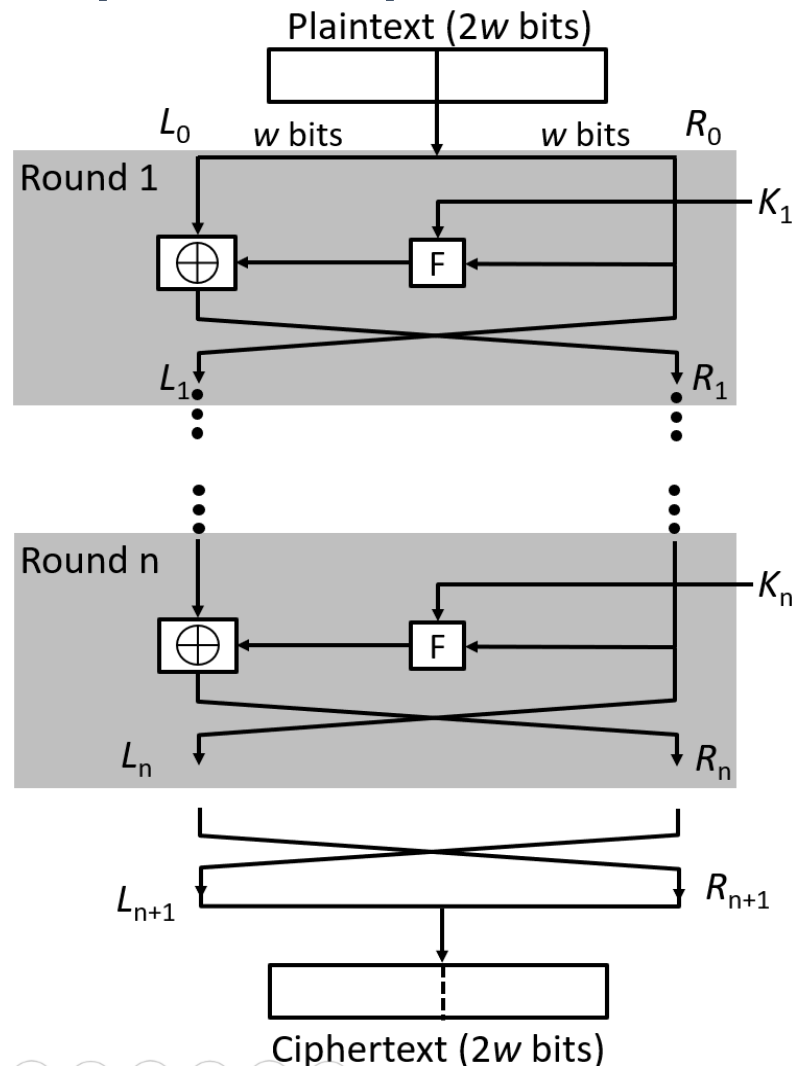
Triple DES

AES

Stream cipher and Block Cipher:



Block Cipher Principle – Feistel Structure



1. Plaintext is split into 32-bit halves L_i and R_i
2. R_i is fed into the function F .
3. The output of function F is then XORed with L_i
4. Left and right half are swapped.

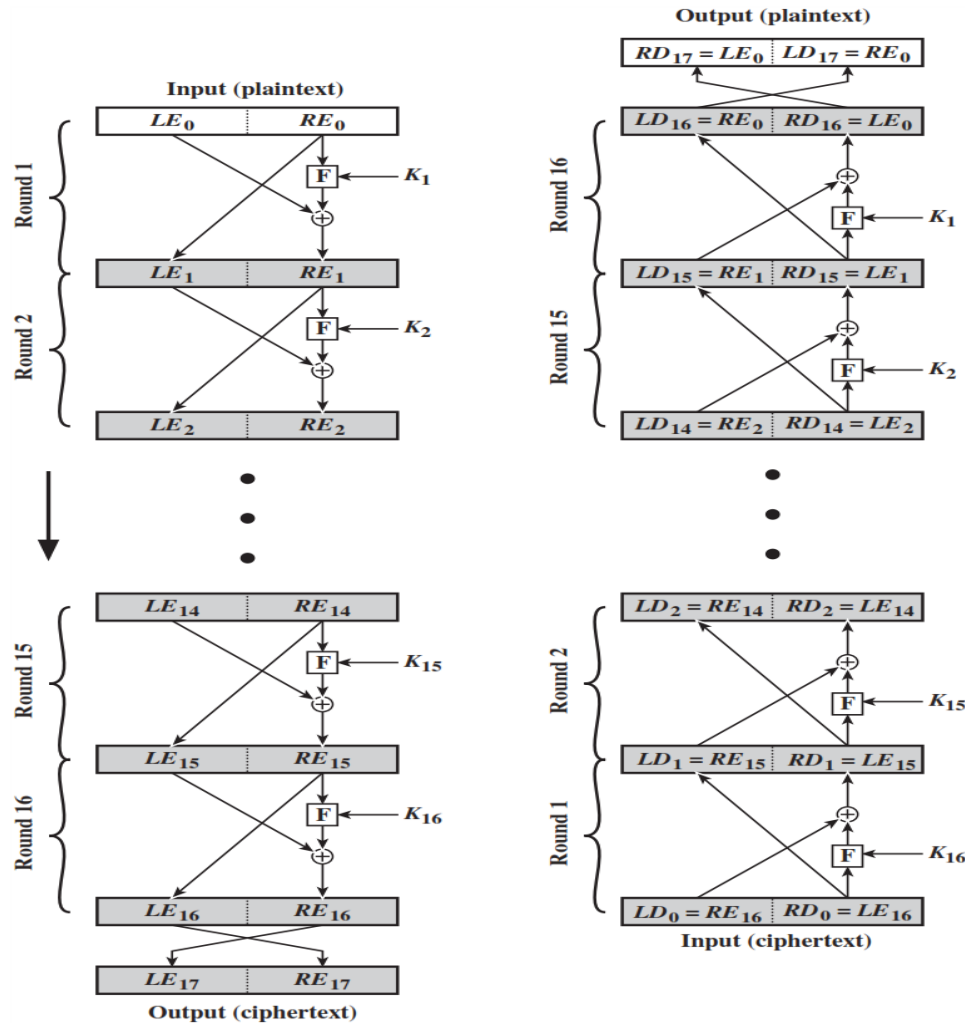
$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

$$L_i = R_{i-1}$$

Block Cipher Principle – Fiestel Structure

1. **Block size:** Common block size of 64-bit. However, the new algorithms uses a 128-bit, 256-bit block size.
2. **Key size:** Key sizes of 64 bits or less are now widely considered to be insufficient, and 128 bits has become a common size.
3. **Number of rounds:** A typical size is 16 rounds.
4. **Round function F:** This phase consisting of sixteen rounds of the same function, which involves both permutation and substitution functions. Again, greater complexity generally means greater resistance to cryptanalysis.
5. **Subkey generation algorithm:** For each of the sixteen rounds, a different subkey (K_i) derived from main key by the combination of a left circular shift and a permutation. Greater complexity in this algorithm should lead to greater difficulty of cryptanalysis.

Block Cipher Principle – Feistel Structure



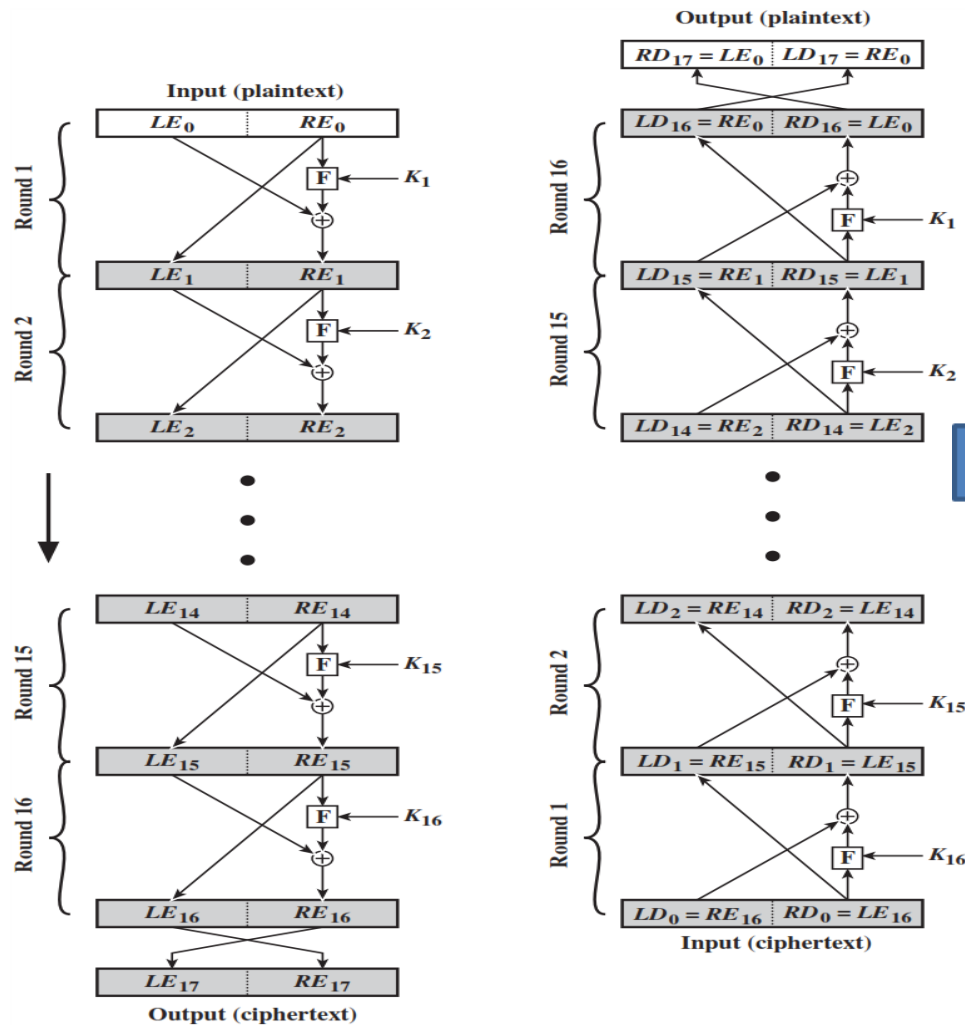
Prove that o/p of first round of Decryption is equal to 32-bit swap o i/p of 16th round of Encryption
 $LD_1 = RE_{15}$ & $RD_1 = LE_{15}$

On Encryption Side:

$$LE_{16} = RE_{15}$$

$$RE_{16} = LE_{15} \oplus F(RE_{15}, K_{16})$$

Block Cipher Principle – Feistel Structure



On Decryption Side:

$$LD_1 = RD_0 = LE_{16} = RE_{15}$$

$$RD_1 = LD_0 \oplus F(RD_0, K_{16})$$

$$= RE_{16} \oplus F(RE_{15}, K_{16})$$

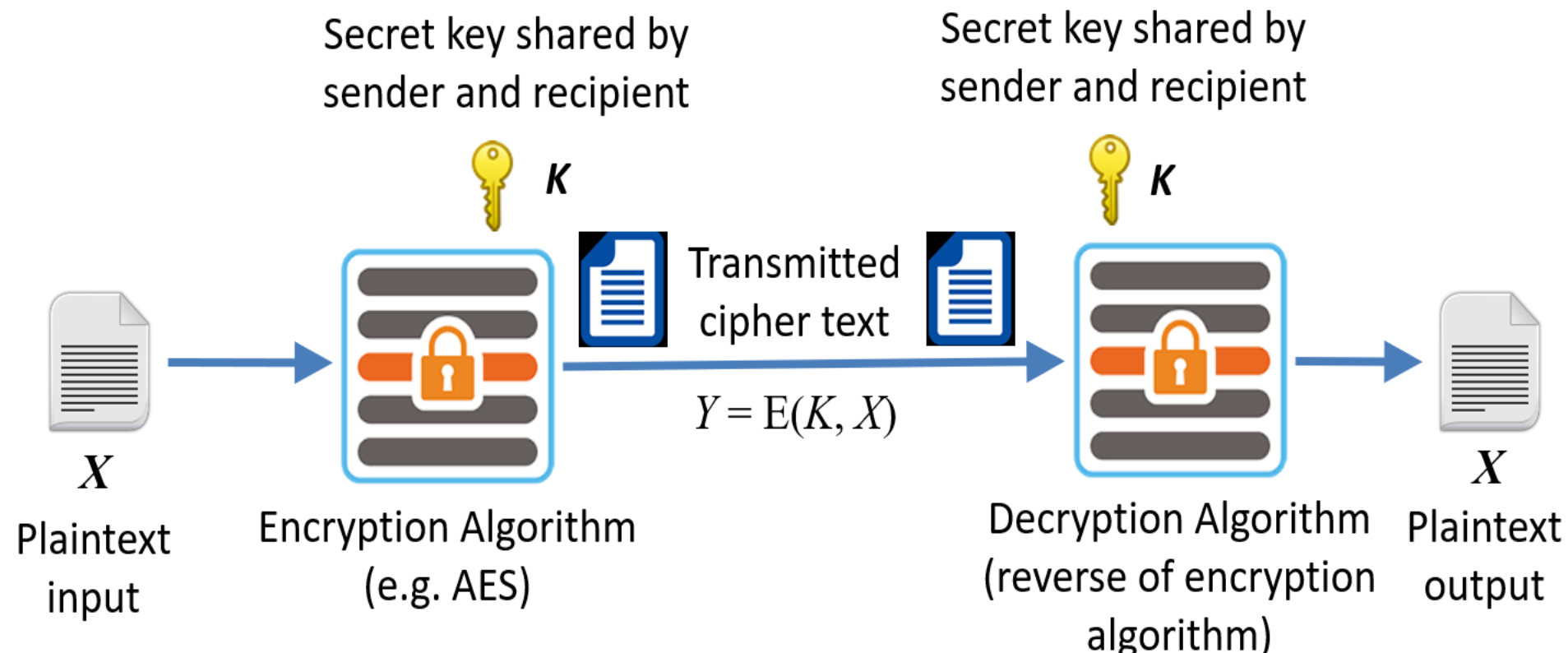
$$= [LE_{15} \oplus F(RE_{15}, K_{16})] \oplus F(RE_{15}, K_{16})$$

Thus,

$$LD_1 = RE_{15} \text{ \& } RD_1 = LE_{15}$$

XOR Associativity Property
 $\because [A \oplus B] \oplus C = A \oplus [B \oplus C]$

Symmetric Cipher Model



Stream cipher and Block Cipher:

A stream cipher is one that encrypts a digital data stream one bit or one byte at a time.

Examples:

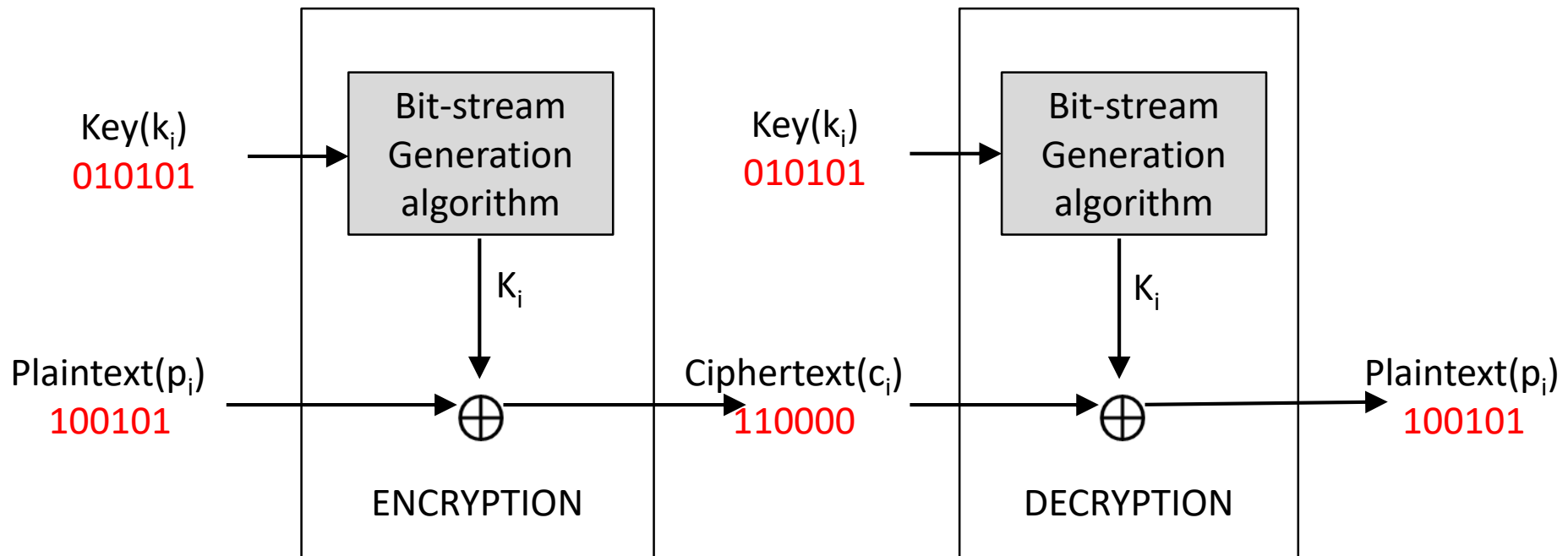
Autokeyed Vigenère cipher

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Stream cipher and Block Cipher:



Stream cipher and Block Cipher:

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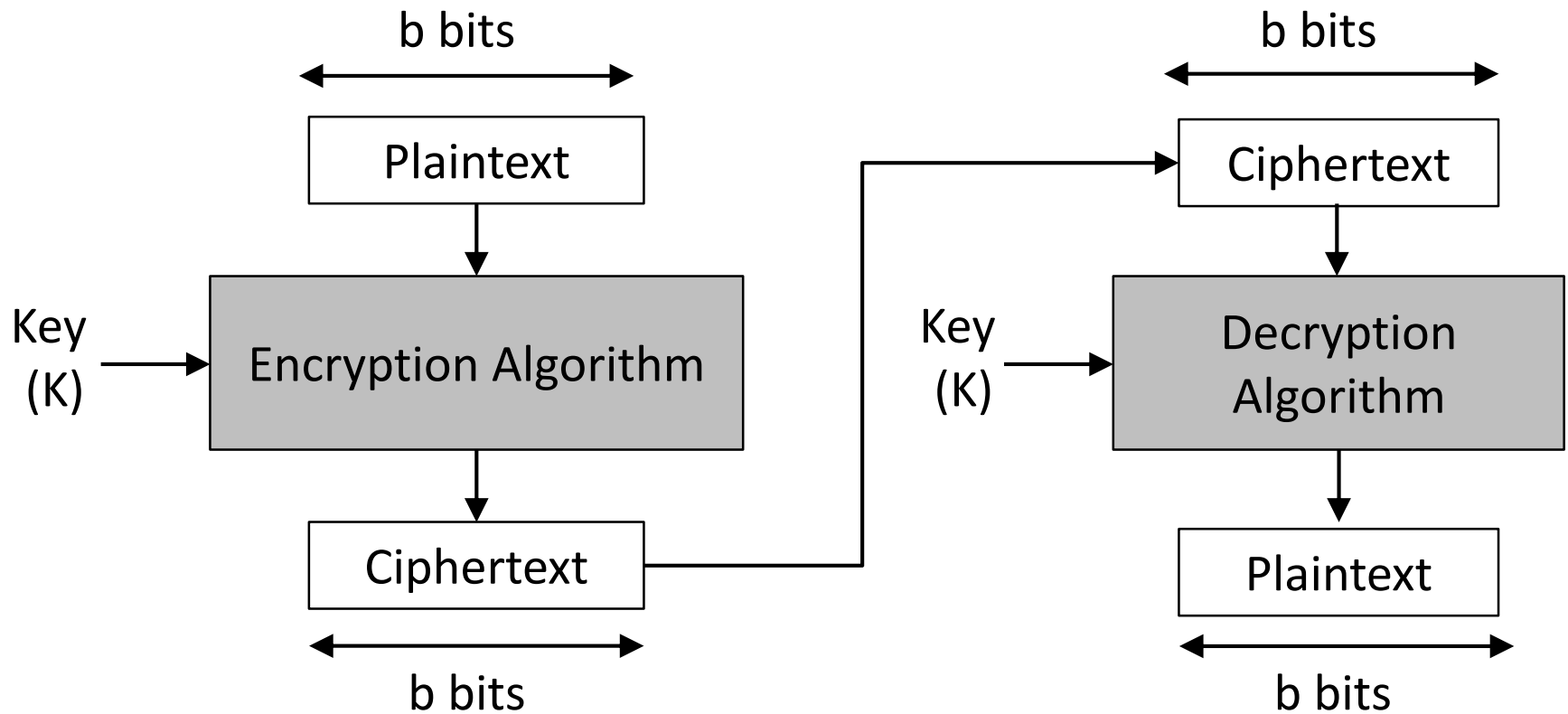
Feistel cipher

DES

Triple DES

AES

Stream cipher and Block Cipher:



Confusion & Diffusion:

Confusion

- Confusion **hides the relationship** between the **cipher text** and the **key**.
- This is achieved by the use of a complex **substitution algorithm**.

Diffusion

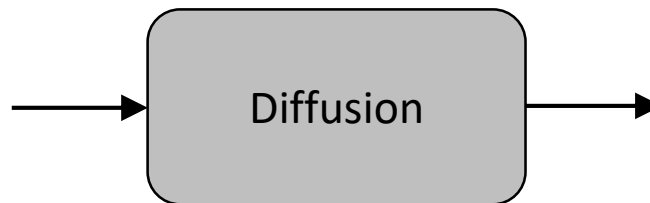
- Diffusion **hides the relationship** between the **cipher text** and the **plaintext**.
- This is achieved by changing **one plaintext digit** which **affect** the value of **many cipher text digits**.

X1=0010 1011

X2=0000 1011

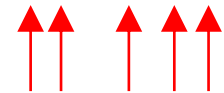


Single bit flip



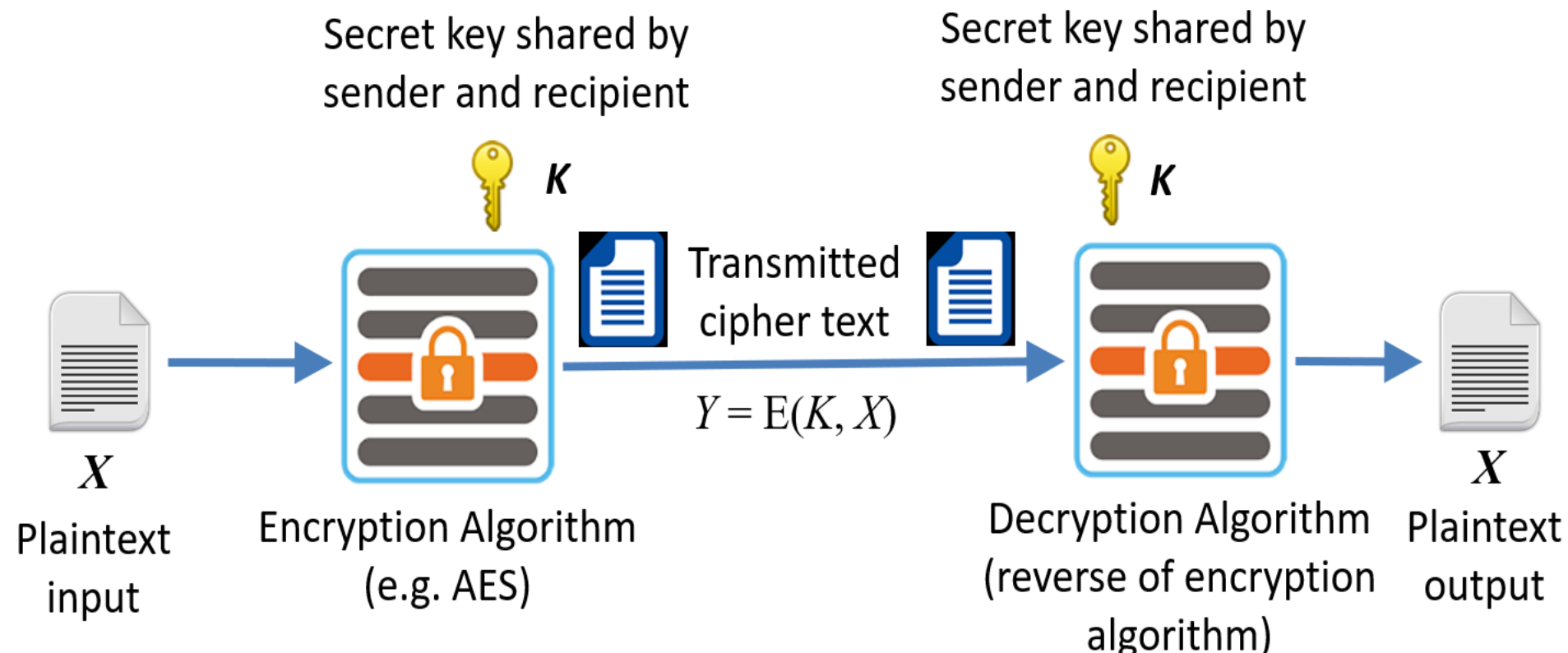
Y1=1011 1001

Y2=0110 1100



Many bit flips

Symmetric Cipher Model



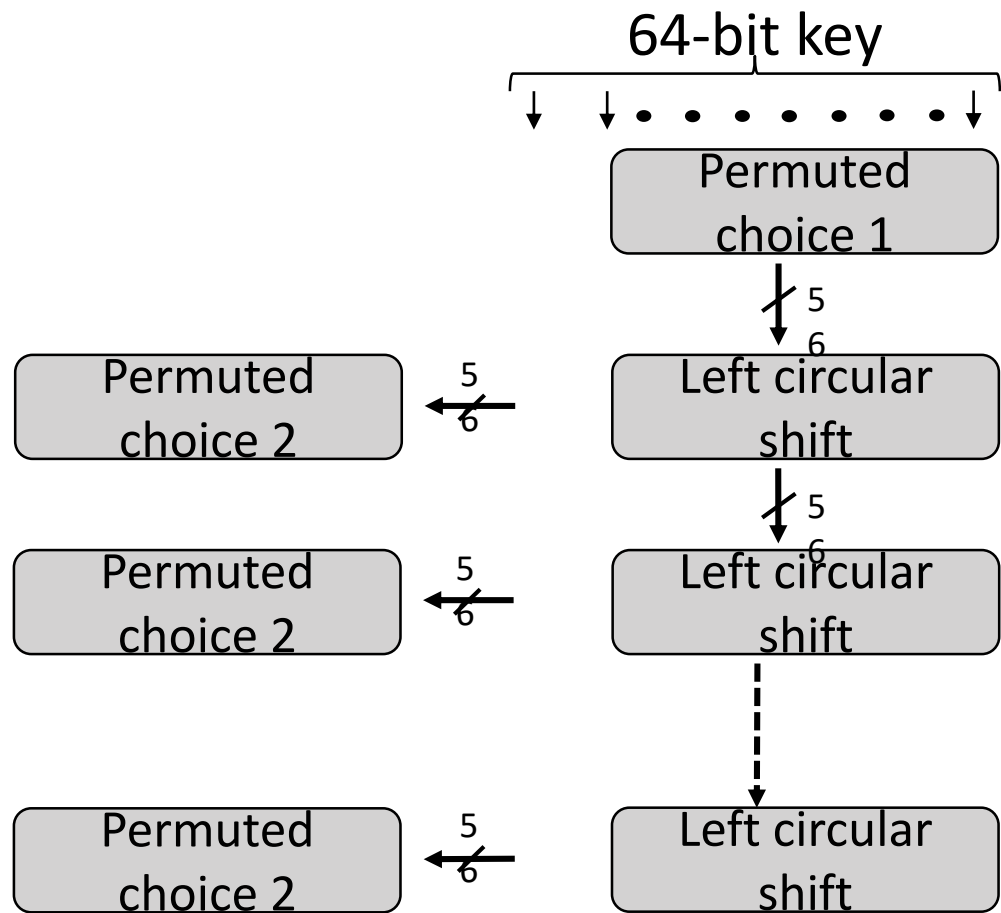
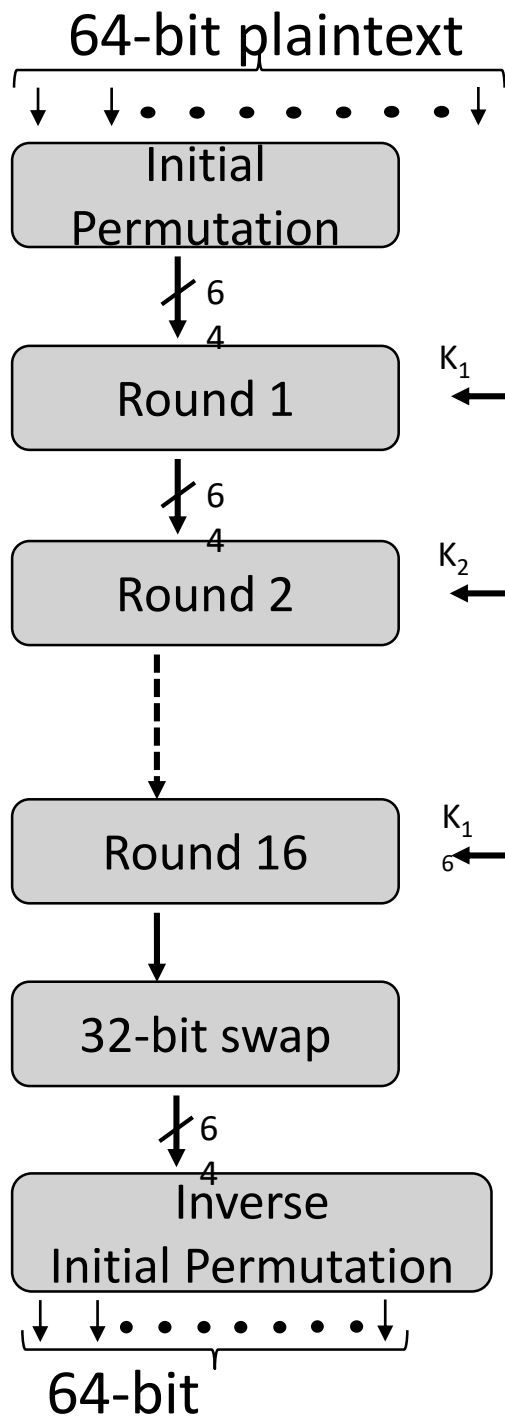
Data Encryption Standard (DES):

Type: Block Cipher

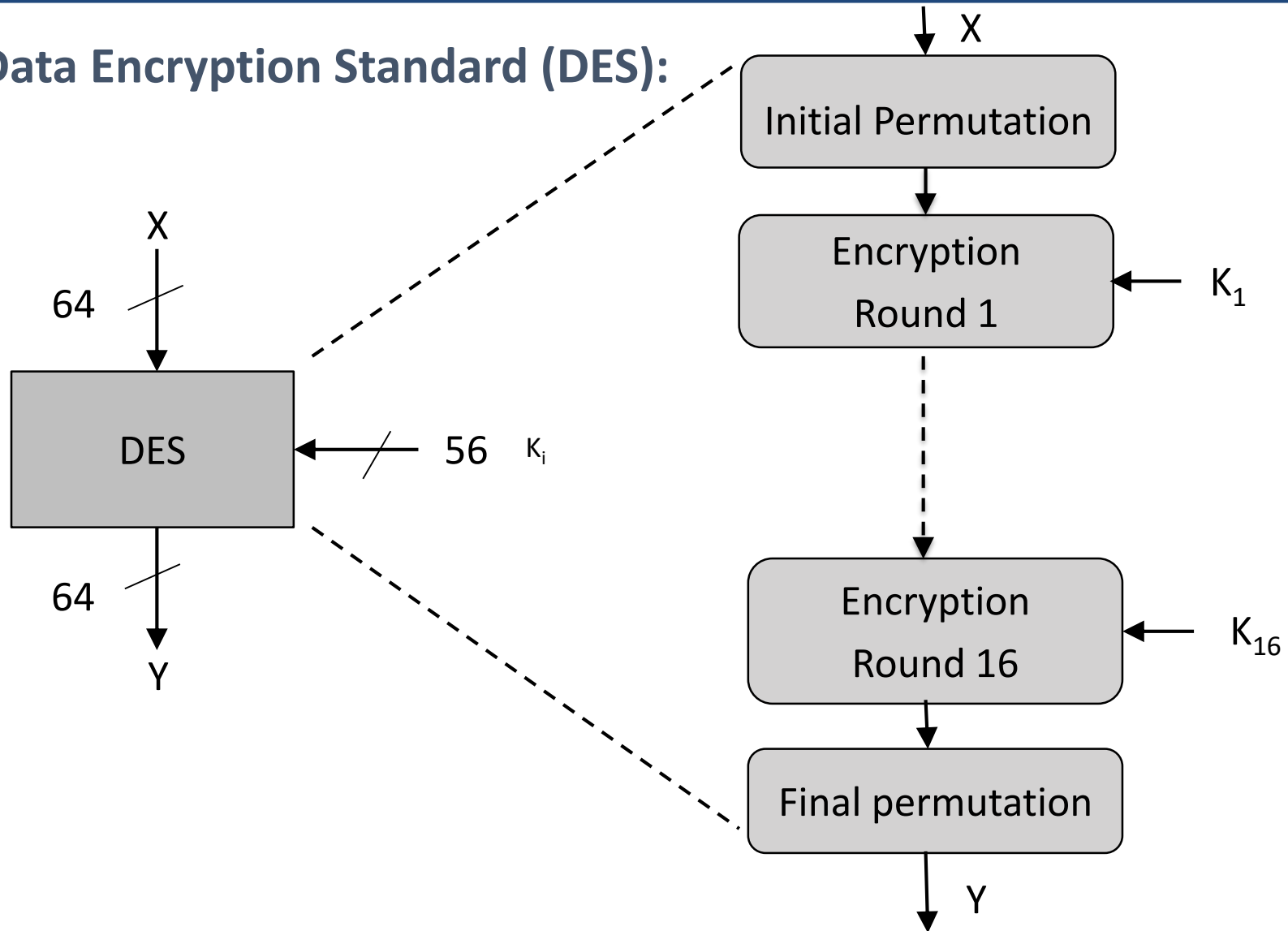
Block Size : 64-bit

Key Size: 64-bit, with only 56-bit effective

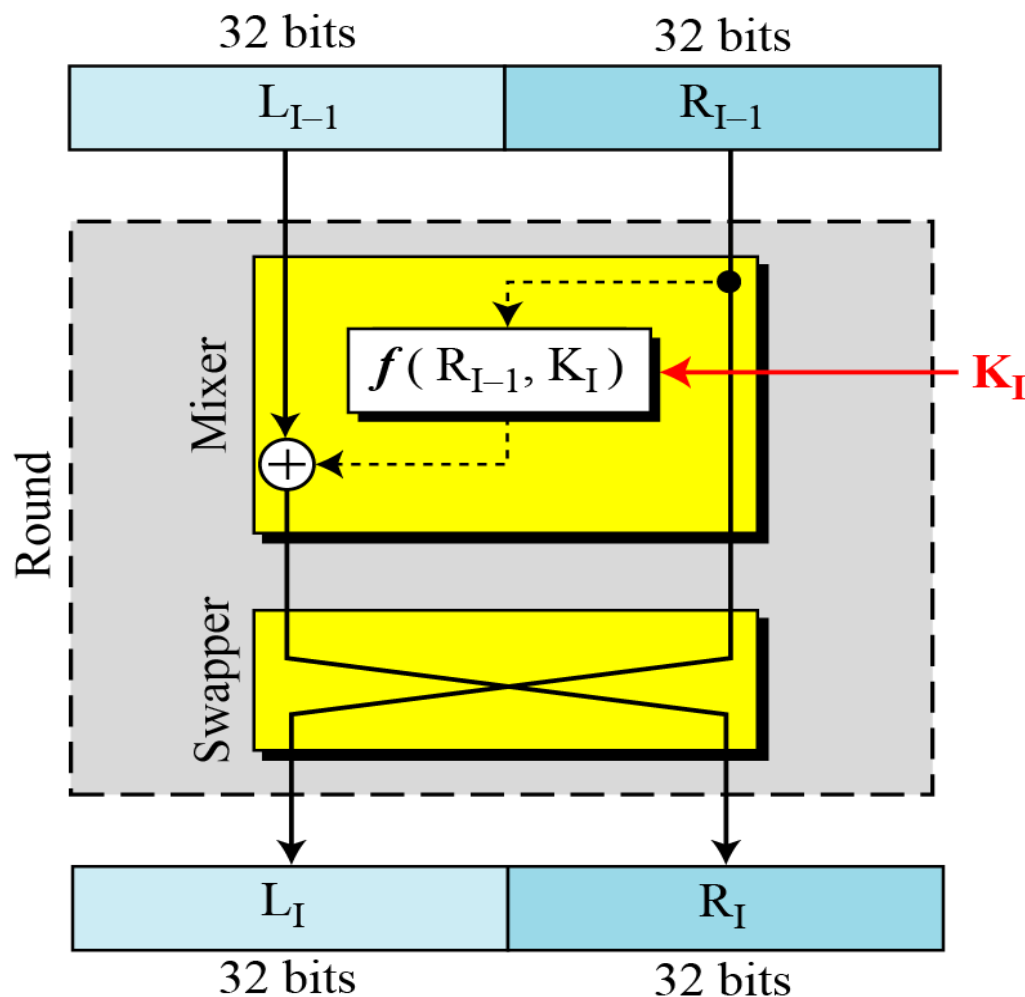
Number of Rounds: 16

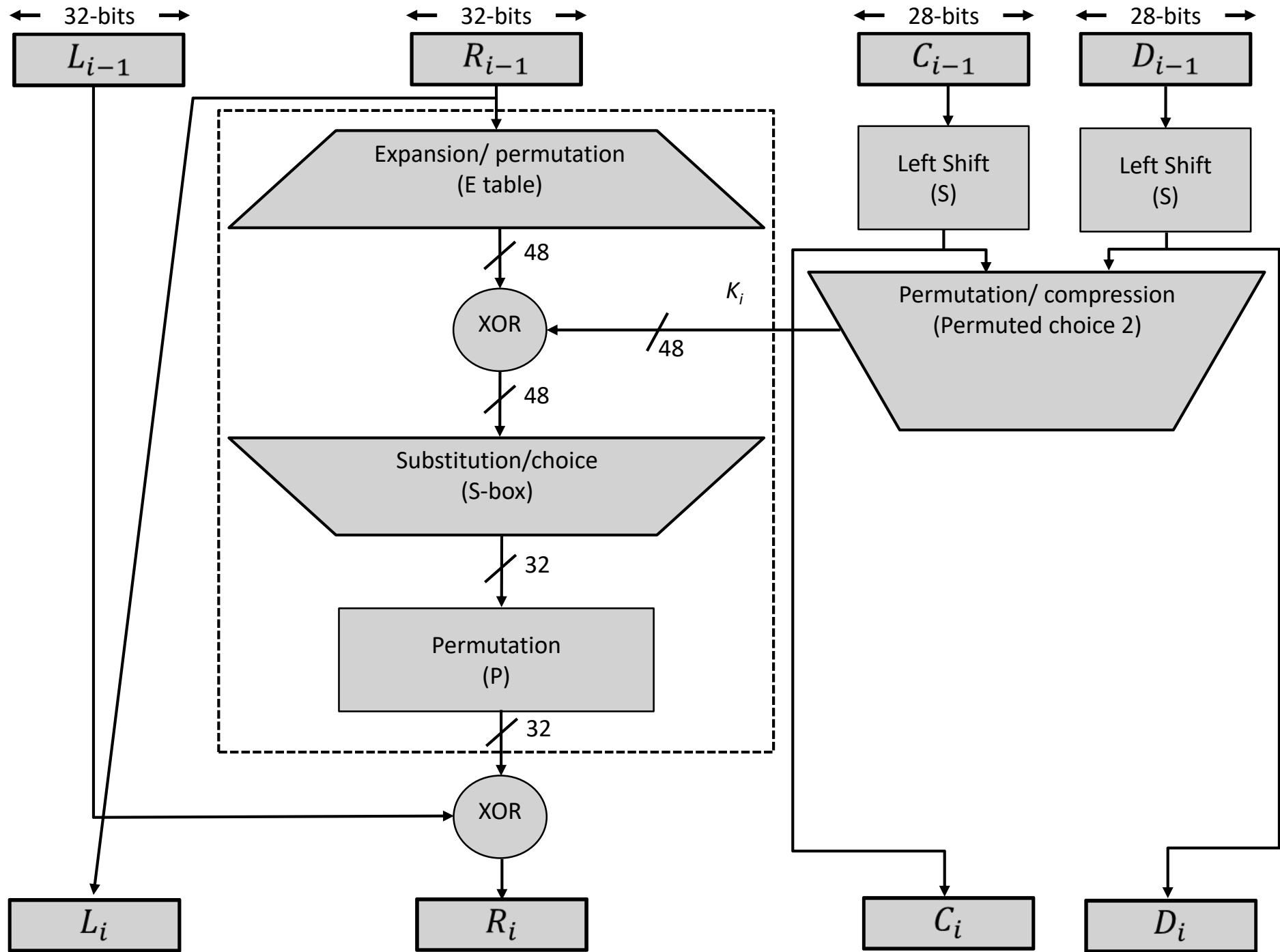


Data Encryption Standard (DES):



Data Encryption Standard (DES) – Single round of DES:



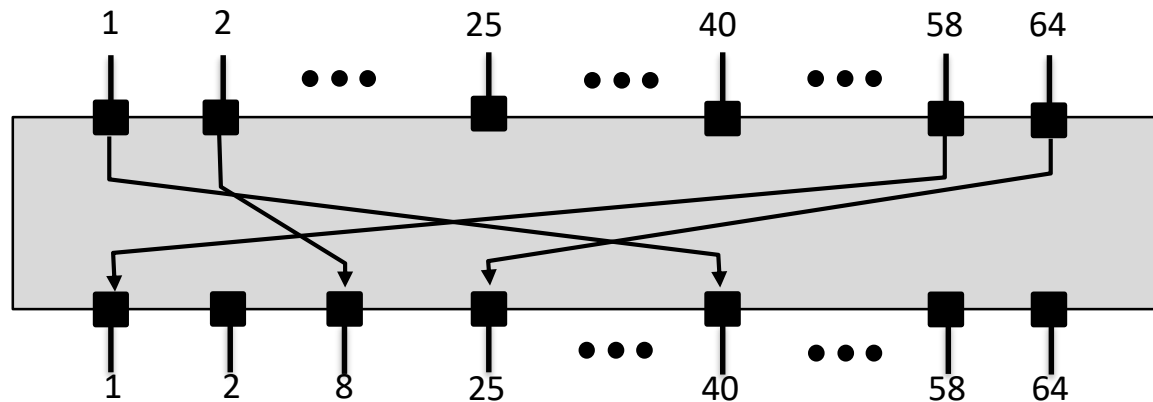


Data Encryption Standard (DES):

1. Initial permutation: First, the 64-bit plaintext passes through an initial permutation (IP) that rearranges the bits to produce the permuted input.
2. The F function: This phase consisting of sixteen rounds of the same function, which involves both permutation and substitution functions.
3. Swap: L and R swapped again at the end of the cipher, i.e., after round 16 followed by a final permutation.
4. Inverse (Final) permutation: It is the inverse of the initial permutation.
5. Subkey generation: For each of the sixteen rounds, a different subkey (K_i) derived from main key by the combination of a left circular shift and a permutation.

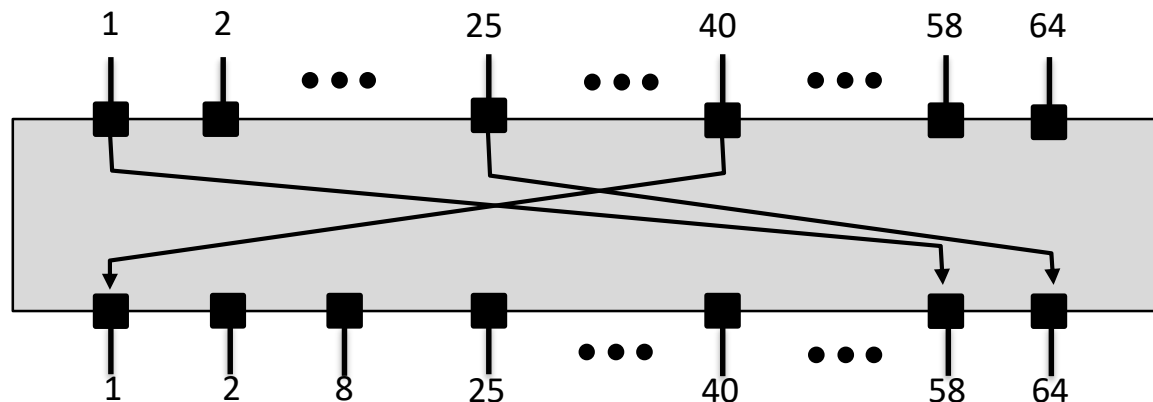
Data Encryption Standard (DES): - Initial Permutation

The initial permutation of the DES algorithm changes the order of the plaintext prior to the first round of encryption



16 Rounds

The final permutation occurs after the sixteen rounds of DES are completed. It is the inverse of the initial permutation.



Data Encryption Standard (DES): Initial and Final Permutation

IP							
58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

IP ⁻¹							
40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

Data Encryption Standard (DES): The f Function

1. Main operation of DES

▪ f-function inputs:

R_{i-1} and round key K_i

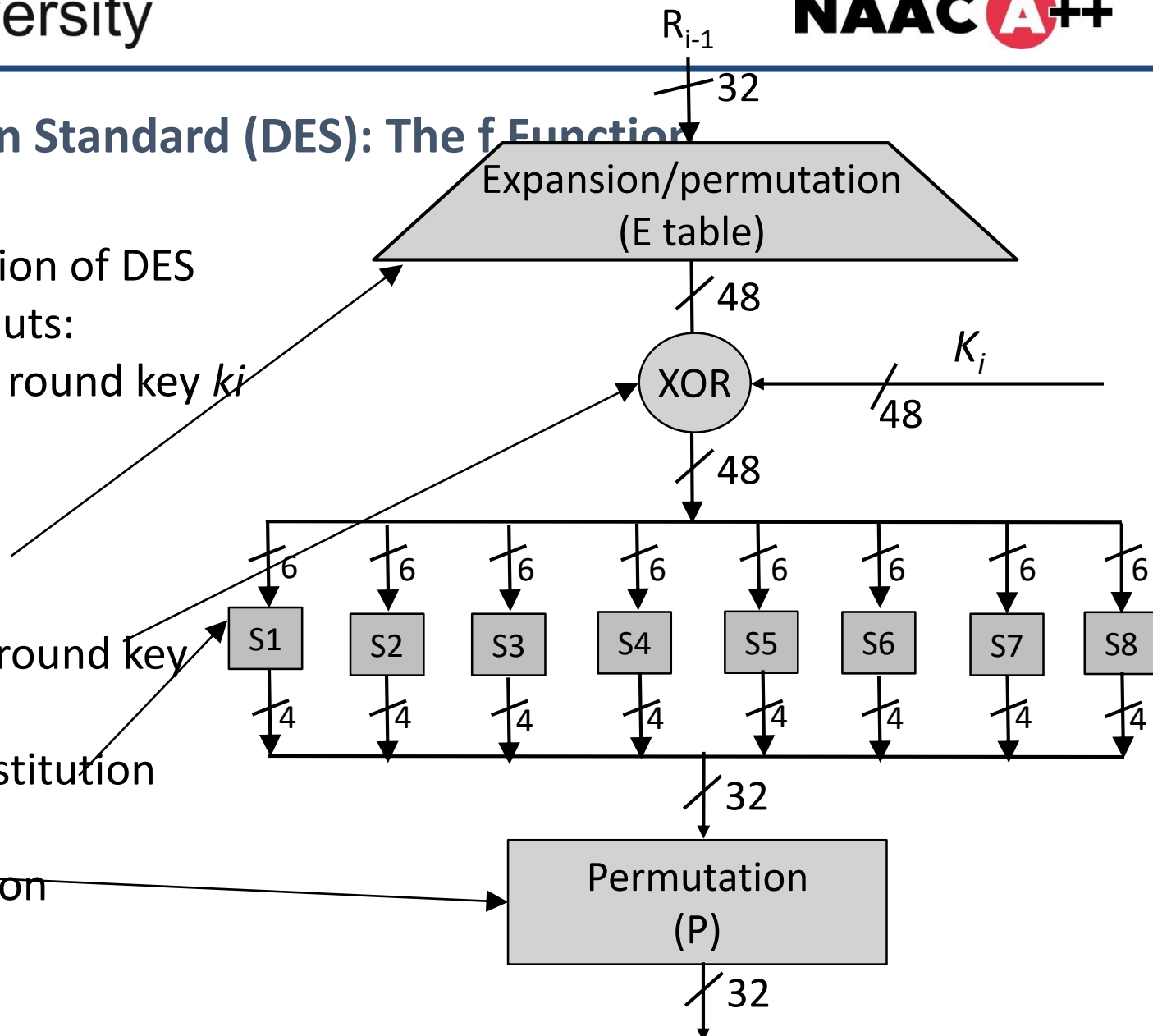
▪ 4 Steps:

1. Expansion E

2. XOR with round key

3. S-box substitution

4. Permutation



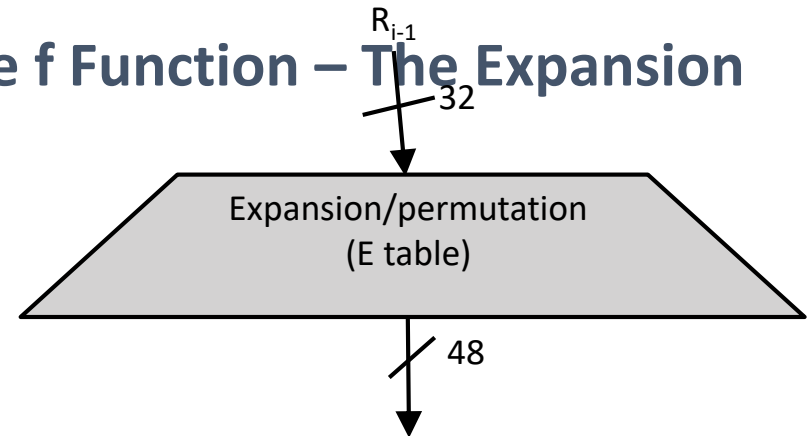
Data Encryption Standard (DES): The f Function – The Expansion Function

Main purpose: **Increases diffusion**

Since R_{i-1} is a 32-bit input and K_i is a 48-bit key, we first need to expand R_{i-1} to 48 bits.

Input: (8 blocks, each of them consisting 4 bits) - 32 bits

Output: (8 blocks, each of them consisting 6 bits) – 48 bits



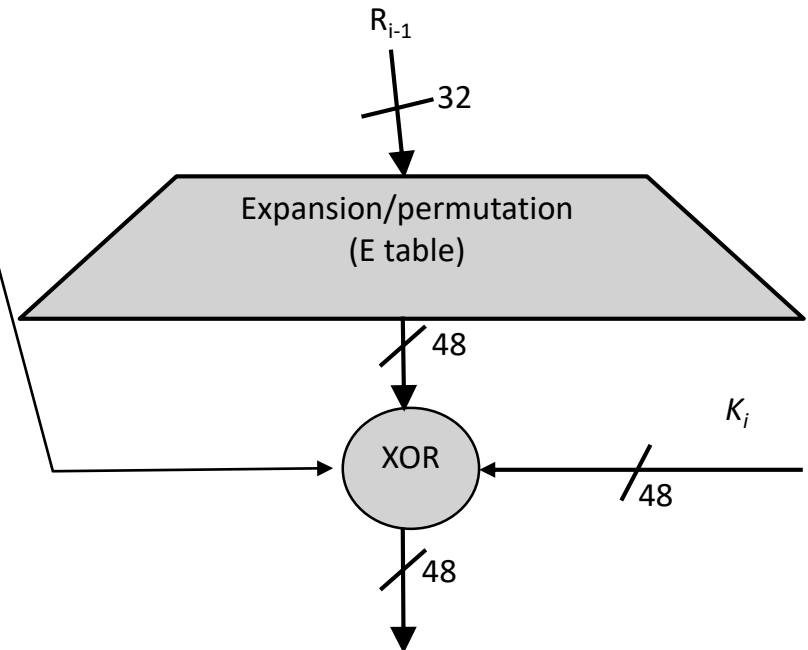
Expansion Table E					
32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

Data Encryption Standard (DES): XOR round Key

XOR Round Key

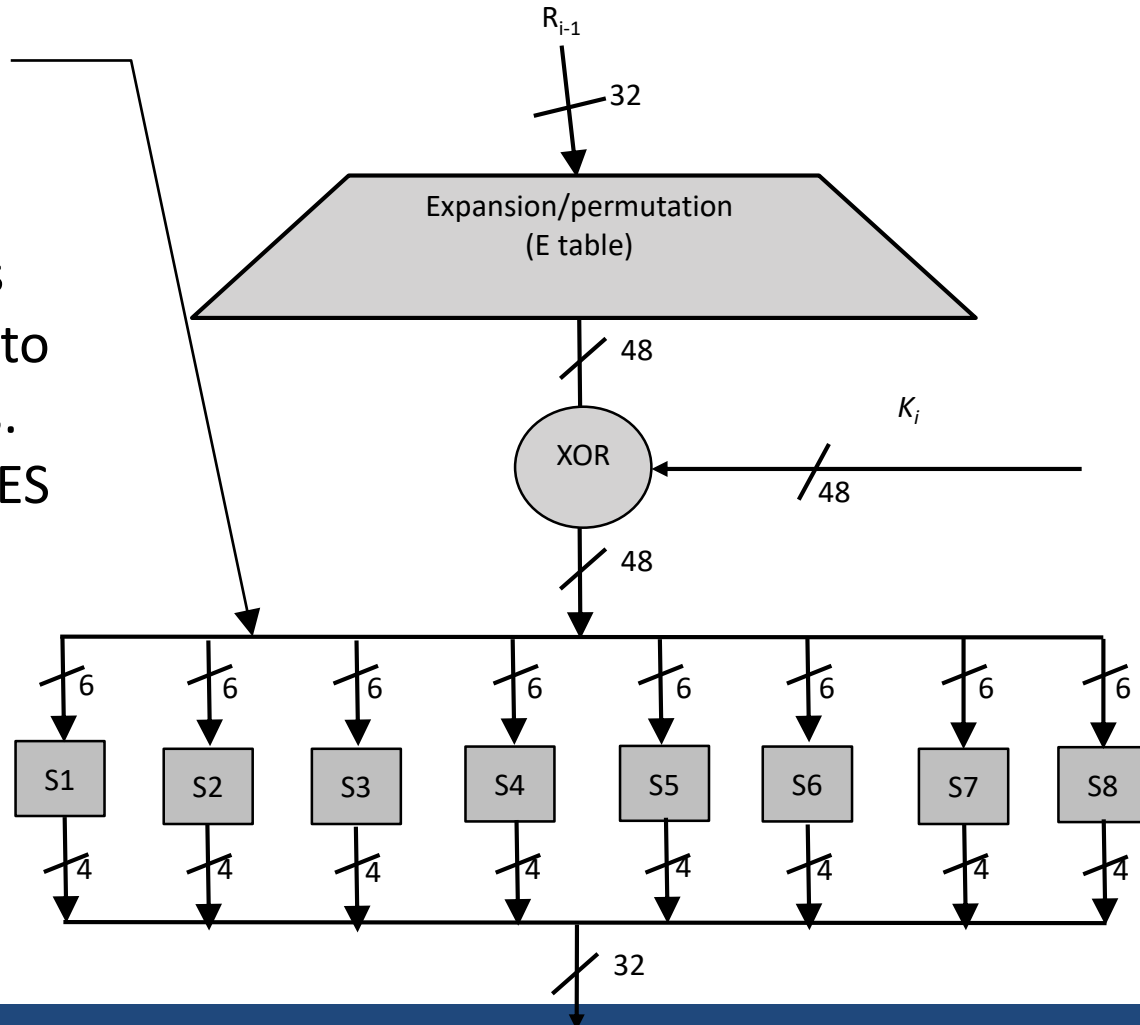
After the expansion permutation, DES uses the XOR operation on the expanded right section and the round key.

Note that both the right section and the key are 48-bits in length now.



Data Encryption Standard (DES): S-Box substitution

- Eight substitution tables.
- 6 bits of input
- 4 bits of output.
- Convert 48 bits to 32 bits
- Non-linear and resistant to differential cryptanalysis.
- Crucial element for DES security!
- Introduces confusion.



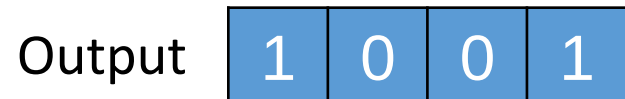
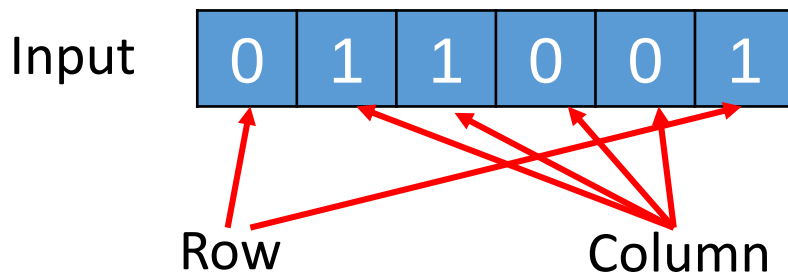
Data Encryption Standard (DES): S-Box substitution

The outer two bits of each group select one row of an S-box.
Inner four bits select one column of an S-box.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	10	03	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

S-box 1

■ Example:

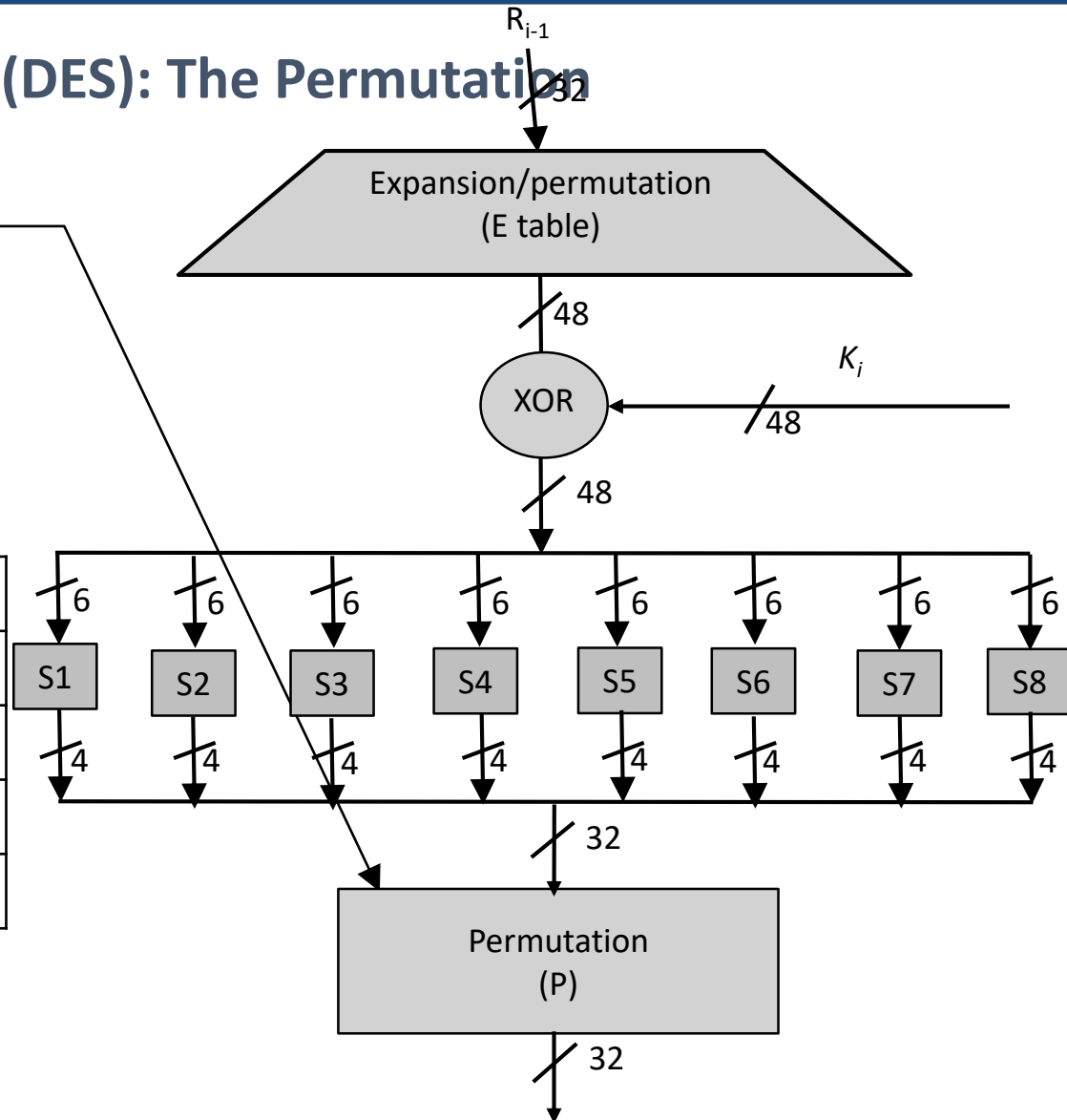


Data Encryption Standard (DES): The Permutation

Permutation P

- Bitwise permutation.
- **Introduces diffusion.**

Permutation Table P							
16	7	20	21	29	12	28	17
01	15	23	26	05	18	31	10
02	08	24	14	32	27	03	09
19	13	30	06	22	11	04	25



Avalanche Effect

Desirable property of any encryption algorithm is that a change in one bit of the plaintext or of the key should produce a change in many bits of cipher text.

DES performs strong avalanche effect.

Plaintext: 0000000000000000	Key: 22234512987ABB23
Ciphertext: 4789FD476E82A5F1	
Plaintext: 0000000000000000 <u>1</u>	Key: 22234512987ABB23
Ciphertext: 0A4ED5C15A63FEA3	

Although the two plaintext blocks differ only in the rightmost bit, the cipher text blocks differ in 29 bits.

This means that changing approximately 1.5 % of the plaintext creates a change of approximately 45 % in the ciphertext.

Strength of DES

The use of 56-bit keys: 56-bit key is used in encryption, there are 256 possible keys. A brute force attack on such number of keys is impractical.

The nature of algorithm: Cryptanalyst can perform cryptanalysis by exploiting the characteristic of DES algorithm but no one has succeeded in finding out the weakness.

Design Principle of Block Cipher :

1. **Confusion** Purpose: Make the relationship between the ciphertext and the encryption key as complex as possible. Achieved by: Using substitution operations (like S-boxes).
Effect: Even a small change in the key or plaintext causes major, unpredictable changes in ciphertext.
2. **Diffusion** Purpose: Spread the influence of a single plaintext bit across many ciphertext bits. Achieved by: Using permutation and mixing operations.
Effect: Changing one bit of the plaintext affects many bits of the ciphertext.

Design Principle of Block Cipher :

3. Kerckhoffs's Principle : A cipher should remain secure even if everything about the system (except the key) is public knowledge. Focuses security entirely on the secrecy of the key, not the algorithm.

4. Iterative Structure (Rounds) Instead of a single operation, block ciphers apply multiple rounds of transformations. Each round improves confusion and diffusion.

Example: AES uses 10, 12, or 14 rounds depending on key size.

5. Key Expansion The key schedule algorithm generates a different subkey for each round from the original key. Strong key expansion ensures better security.

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Multiple Encryptions and Triple DES

Chapter-4: Multiple Encryptions and Triple DES

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Assistant Professor

Department of Computer Science and Engineering

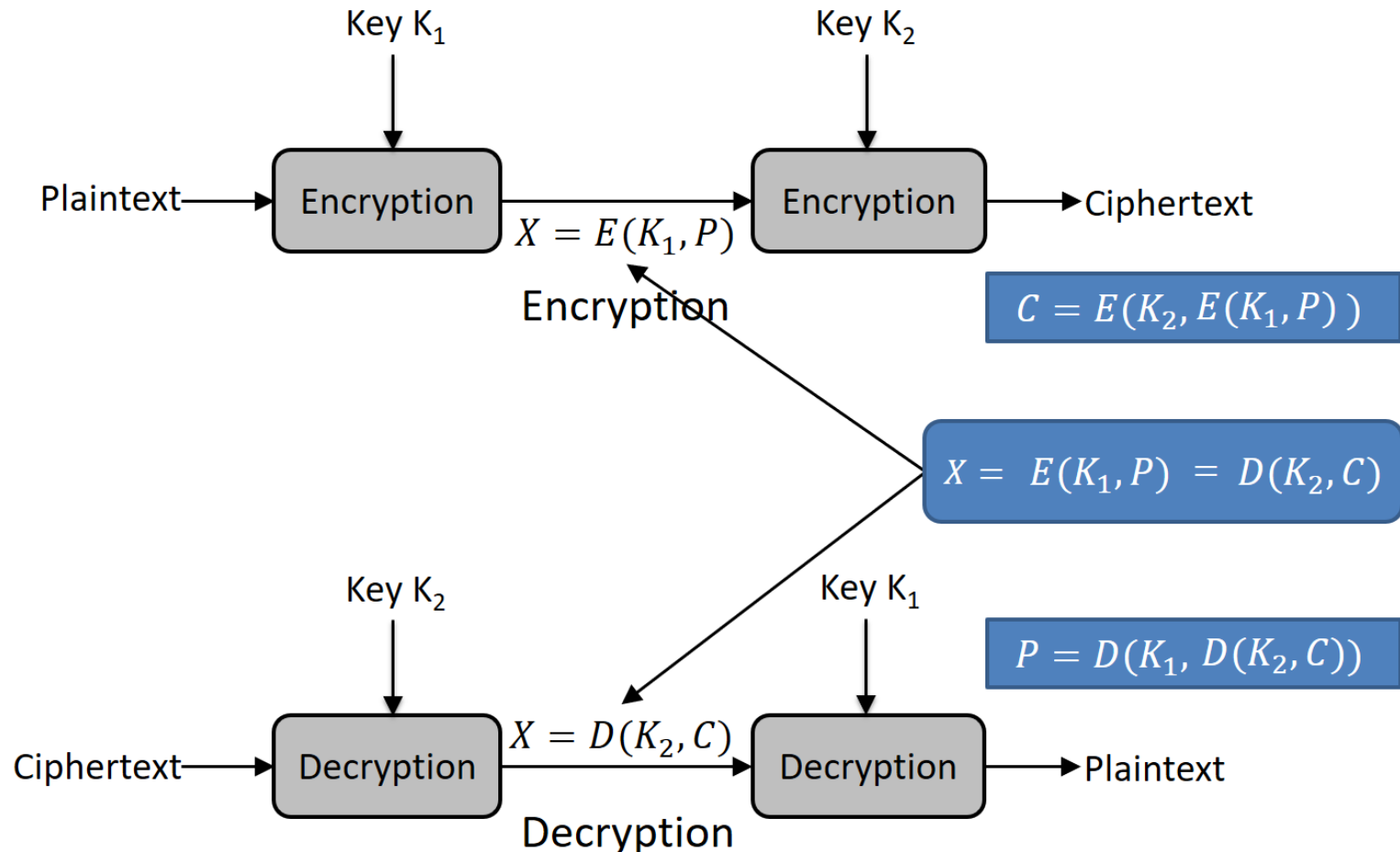
Content

1. Multiple encryptions and triple DES
2. Electronic Code Book
3. Cipher Block Chaining Mode
4. Cipher Feedback mode
5. Output Feedback mode
6. Counter mode

Multiple Encryption

- Given the potential vulnerability of DES to a brute-force attack, there has been considerable interest in finding an alternative.
- For DES requires 256 operations for brute force attack.
- One approach is to design a completely new algorithm, of which AES is a prime example.
- Another alternative, which would preserve the existing investment in software and equipment, is to use multiple encryption with DES and multiple keys.

Double DES

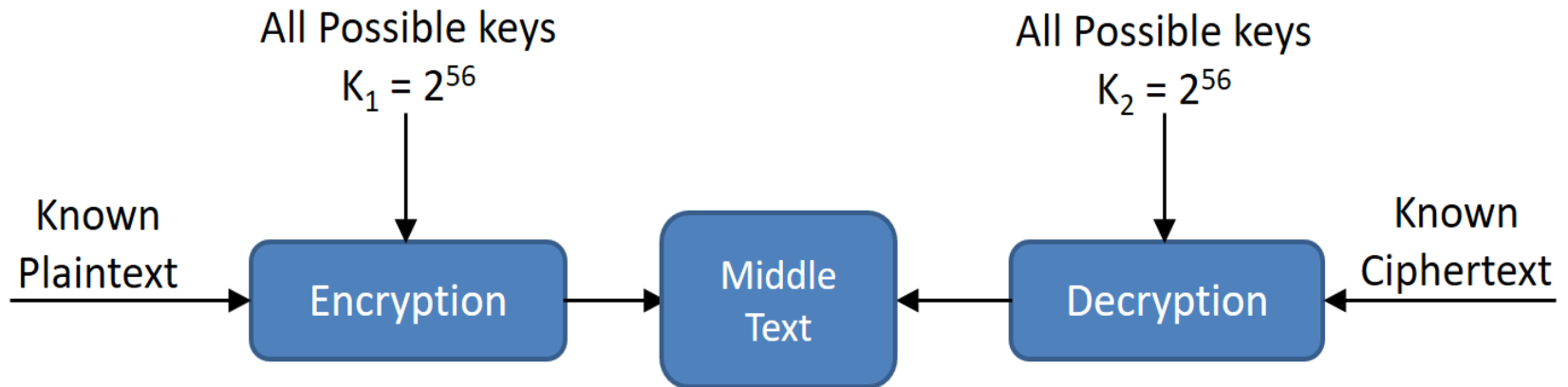


Meet in the Middle Attack:

This attack involves encryption from one end, decryption from the other and matching the results in the middle.

Suppose cryptanalyst knows P_i and corresponding C_i .

Now, the aim is to obtain the values of K_1 and K_2 .

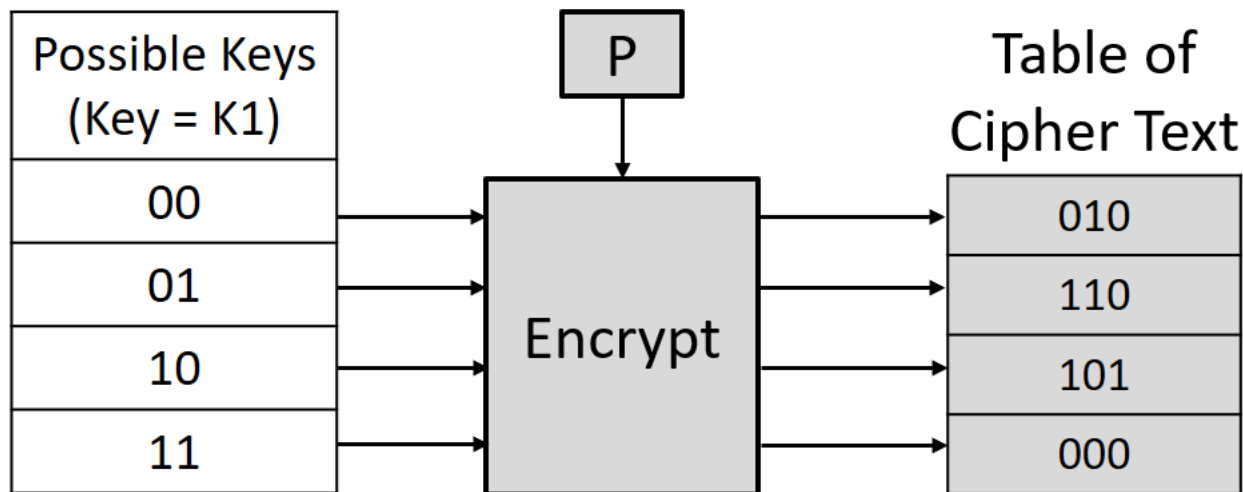


No. of Encryptions and Decryptions: $256 + 256 = 257$

For Double DES requires 257 operations for brute force attack.

Meet in the Middle Attack:

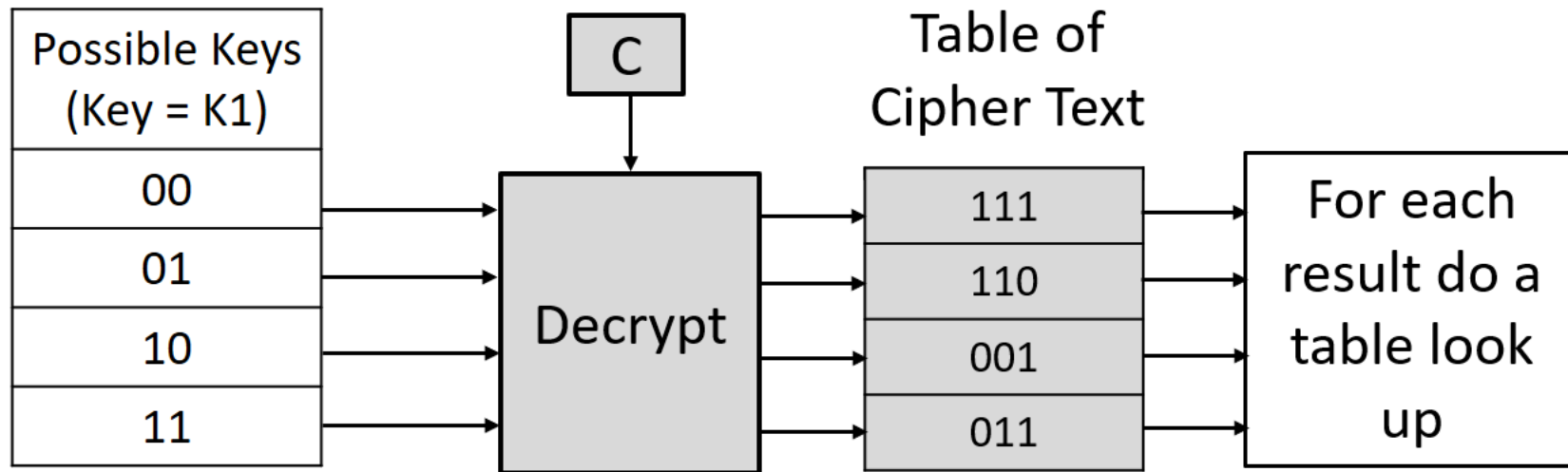
- For all possible values (256) of key K1, the cryptanalyst would encrypt the known plaintext by performing $E(K1, P)$.
- The cryptanalyst would store output in a table.



Cryptanalyst encryption operation

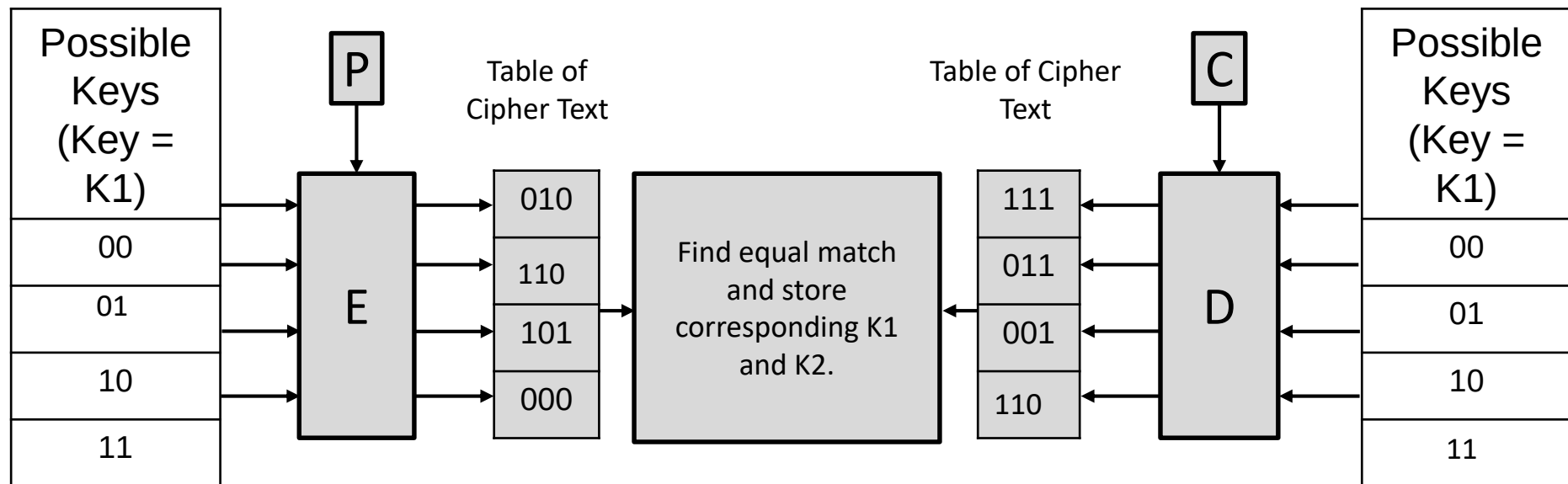
Meet in the Middle Attack:

- Cryptanalyst decrypt the known ciphertext with all possible values of K_2 .
- In each case cryptanalyst will compare the resulting value with the all values in the table of ciphertext.



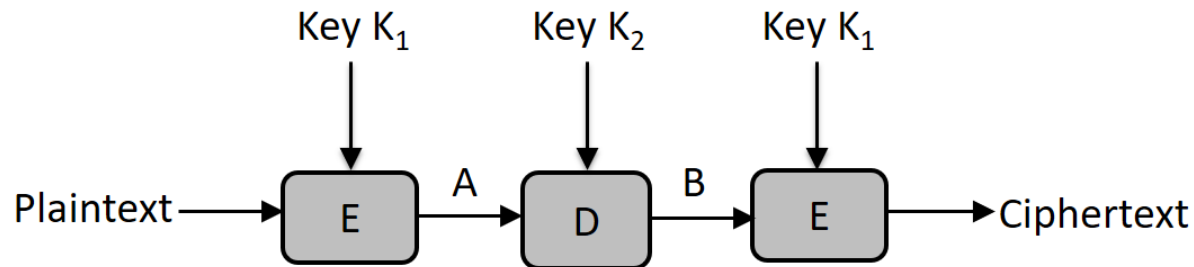
Cryptanalyst decryption operation

Meet in the Middle Attack:

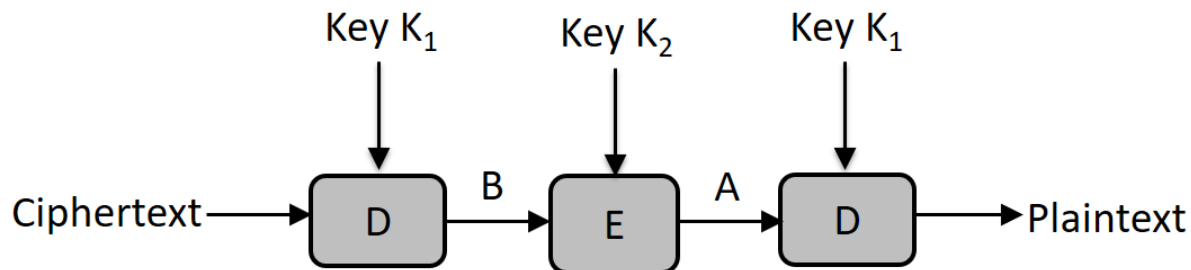


Values of K1=01 and K2=11

Triple DES



$$C = E(K_1, D(K_2, E(K_1, P)))$$



$$P = D(K_1, E(K_2, D(K_1, C)))$$

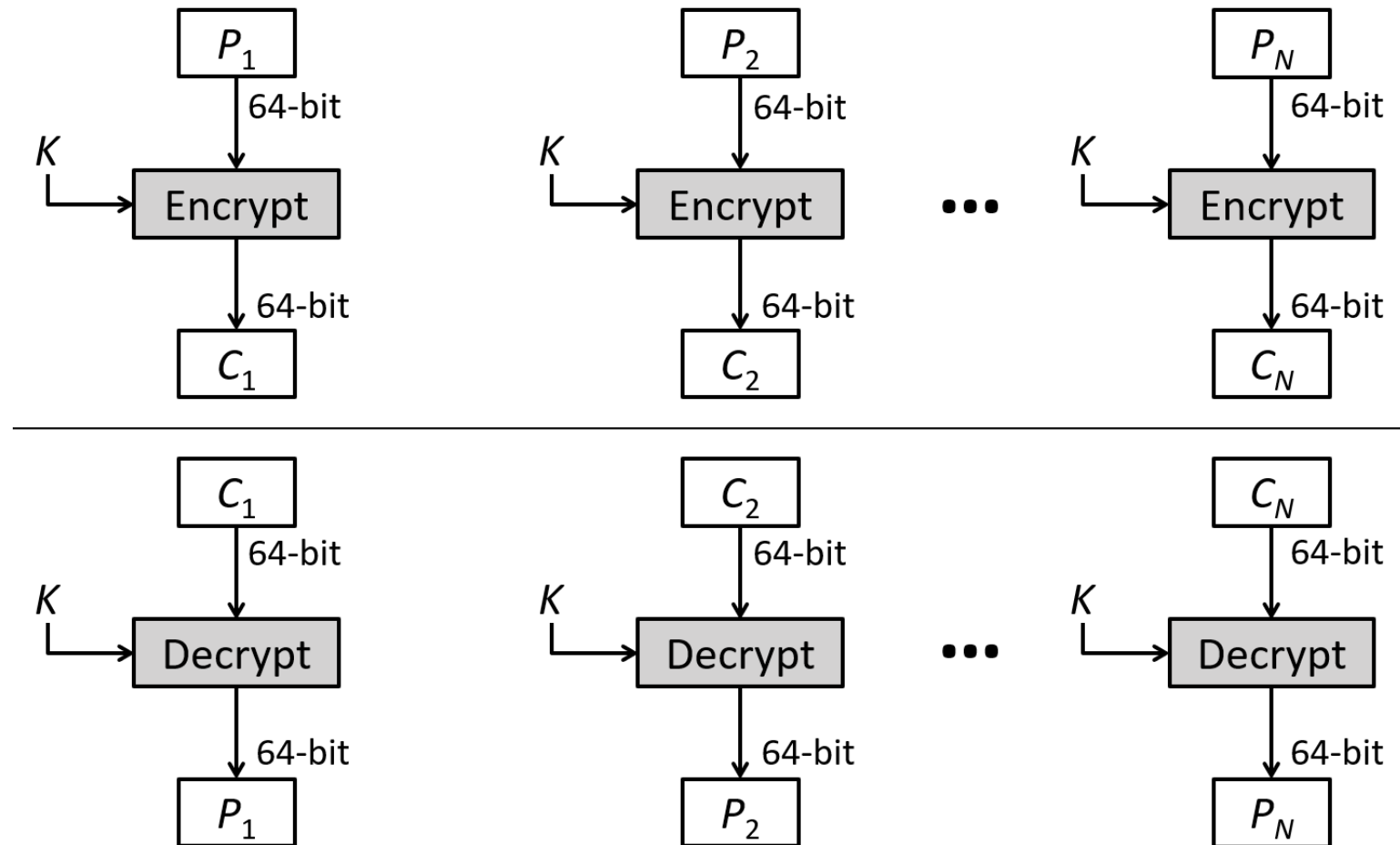
Block Cipher Modes of Operation

- To apply a block cipher in a variety of applications, five "modes of operation" have been defined.
 1. Electronic Code Book (ECB)
 2. Cipher Block Chaining (CBC)
 3. Cipher Feedback (CFB)
 4. Output Feedback (OFB)
 5. Counter (CTR)
- The five modes are intended to cover a wide variety of applications of encryption for which a block cipher could be used.
- These modes are intended for use with any symmetric block cipher, including triple DES and AES.

Block Cipher Modes of Operation

- Block cipher: operates on fixed length b -bit input to produce b -bit ciphertext.
- What about encrypting plaintext longer than b bits?
- Break plaintext into b -bit blocks (padding if necessary) and apply cipher on each block.

Electronic Code Book - ECB



Electronic Code Book - ECB

- In ECB Mode Plaintext handled one block at a time and each block of plaintext is encrypted using the same key.
- The term codebook is used because, for a given key, there is a unique ciphertext for every b-bit block of plaintext.

$$C_j = E(K, P_j) \quad j = 1, \dots, N$$

$$P_j = D(K, C_j) \quad j = 1, \dots, N$$

Electronic Code Book - ECB

- ECB Advantages:
- No block synchronization between sender and receiver is required.
- OK if some blocks are lost in transit.
- Bit errors caused by noisy channels only affect the corresponding block but not succeeding blocks.
- Block cipher operating can be parallelized.

Electronic Code Book - ECB

- ECB Disadvantages:
- Identical plaintexts result in identical ciphertexts.
- An attacker recognizes if the same message has been sent twice simply by looking at the ciphertext.
- Plaintext blocks are encrypted independently of previous blocks.
- An attacker may reorder ciphertext blocks which results in valid plaintext.

Electronic Code Book - ECB

- Consider an electronic bank transfer

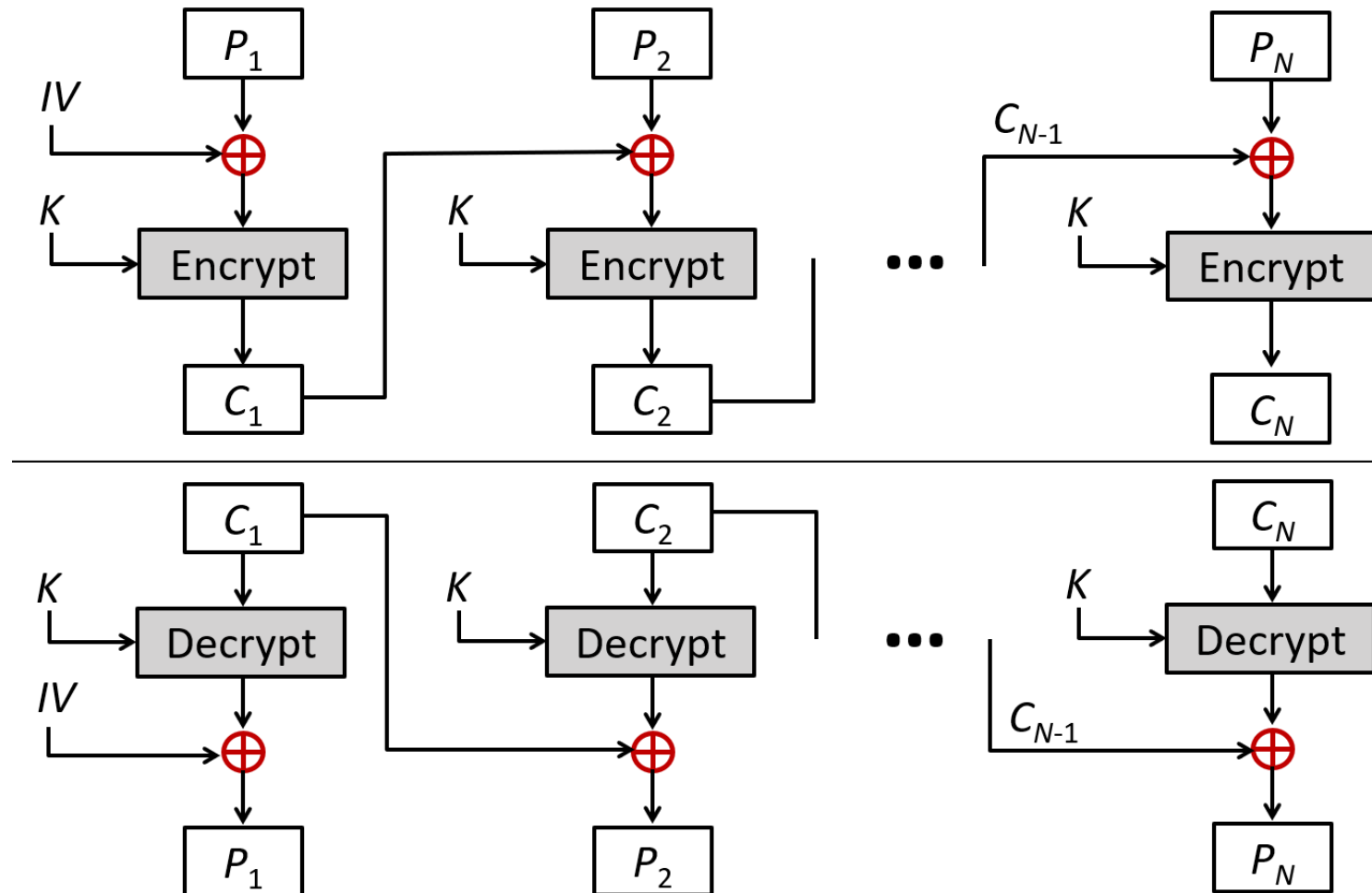
1	2	3	4	5
Sending Bank A	Sending Account #	Receiving Bank B	Receiving Account #	Amount \$

- The attacker sends \$1.00 transfers from his account at bank A to his account at bank B repeatedly.
- He can check for ciphertext blocks that repeat, and he stores blocks 1,3 and 4 of these transfers.
- He now simply replaces block 4 of other transfers with the block 4 that he stored before.
- All transfers from some account of bank A to some account of bank B are redirected to go into the attacker's B account

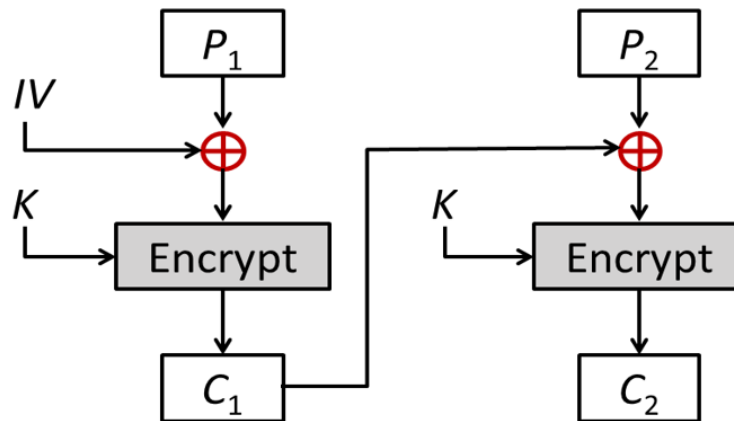
Electronic Code Book - ECB

- Strength: it's simple.
- Weakness:
- Problem: with long message, repetition in plaintext may cause repetition in ciphertext.
- Typical application:
- Secure transmission of short pieces of information (e.g. a temporary encryption key).

Cipher Block Chaining Mode

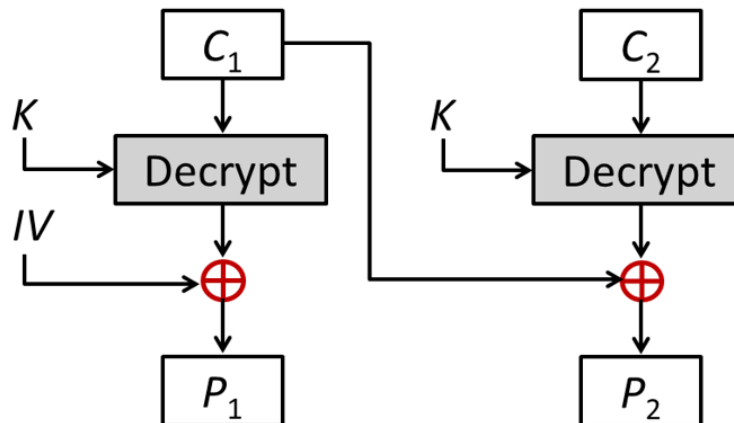


Cipher Block Chaining Mode



$$C_1 = E(K, (P_1 \oplus IV))$$

$$C_j = E(K, (P_j \oplus C_{j-1})) \quad j = 2, \dots, N$$



$$P_1 = D(K, C_1) \oplus IV$$

$$P_j = D(K, C_j) \oplus C_{j-1} \quad j = 2, \dots, N$$

Cipher Block Chaining Mode

CBC is a technique in which the same plaintext block, if repeated, produces different ciphertext blocks.

In this scheme, the input to the encryption algorithm is the XOR of the current plaintext block and the preceding ciphertext block and the same key is used for each block.

To produce the first block of ciphertext, an initialization vector (IV) is XORed with the first block of plaintext.

Cipher Block Chaining Mode

Initialization Vector (IV) must be known by sender/receiver, but it should be kept secret from attacker.

On decryption, the IV is XORed with the output of the decryption algorithm to recover the first block of plaintext.

Cipher Block Chaining Mode

Strength: because of the chaining mechanism of CBC, it is an appropriate mode for encrypting messages of length greater than b bits.

Typical application:

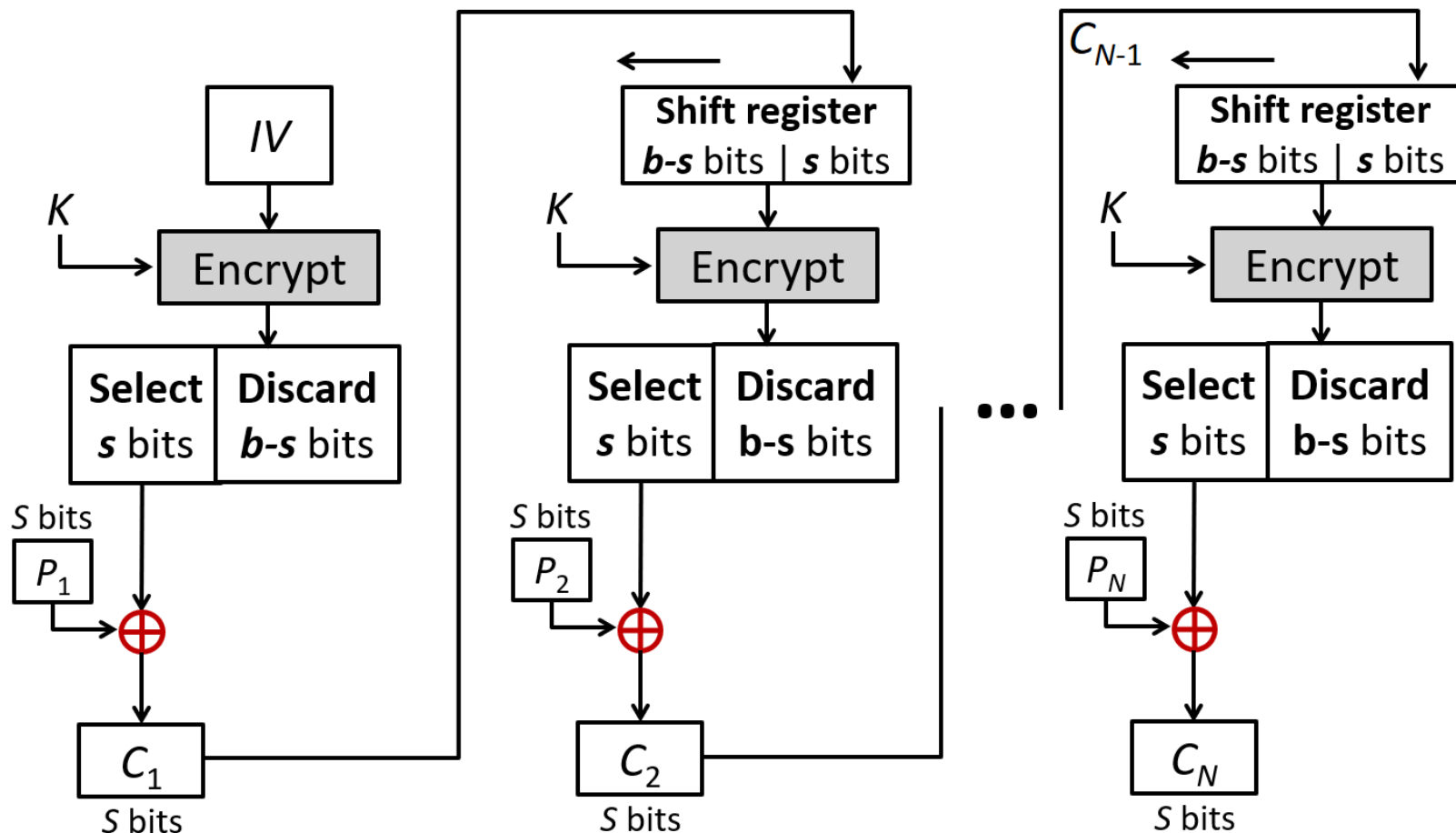
General-purpose block oriented transmission

Authentication

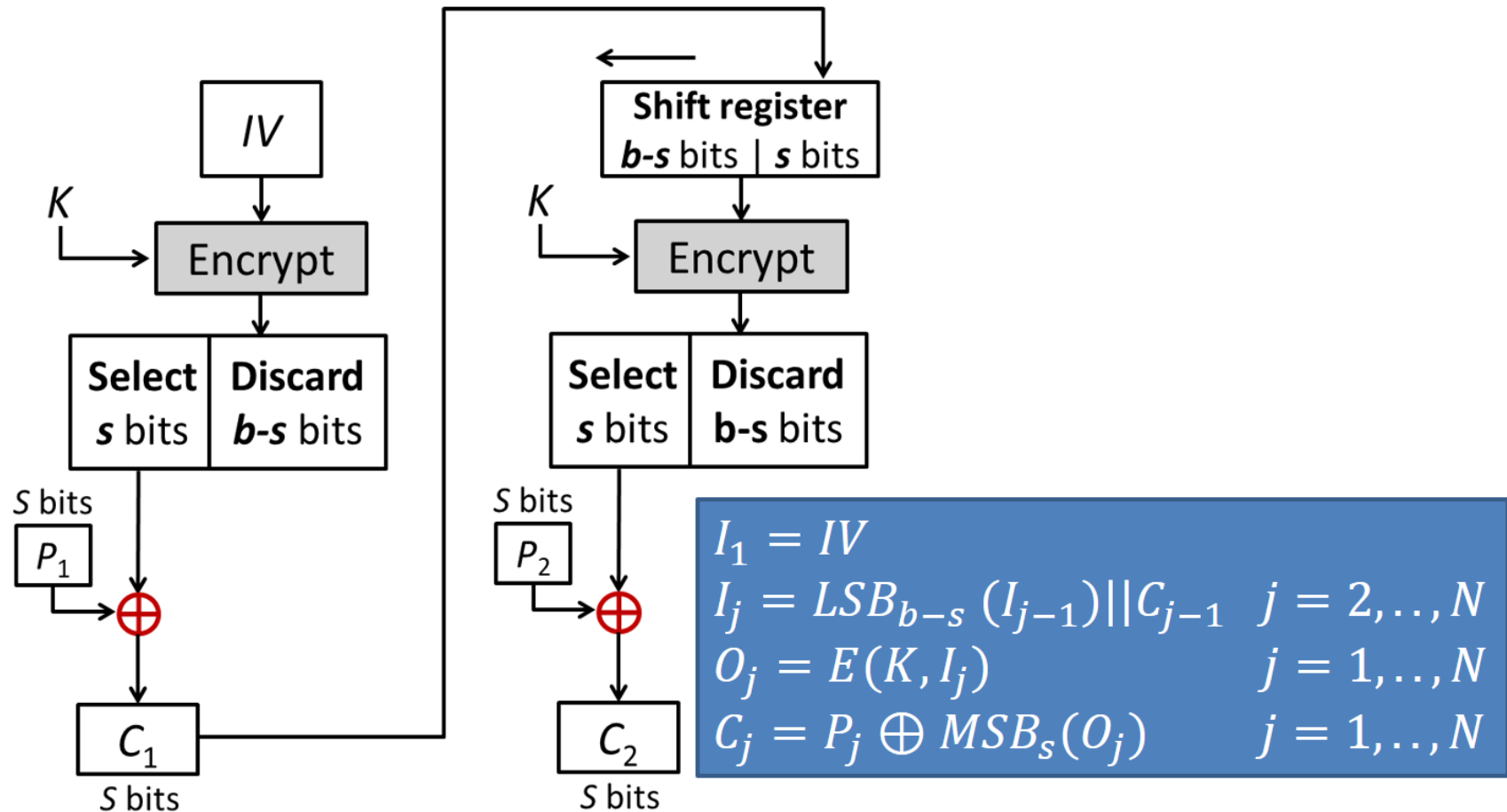
Cipher Feedback Mode:

- For AES, DES, or any block cipher, encryption is performed on a block of b bits. In DES, $b = 64$ and in AES, $b = 128$.
- However, it is possible to convert a block cipher into a stream cipher, using cipher feedback (CFB) mode, output feedback (OFB) mode, and counter (CTR) mode.
- A stream cipher eliminates the need to pad a message to be an integral number of blocks.

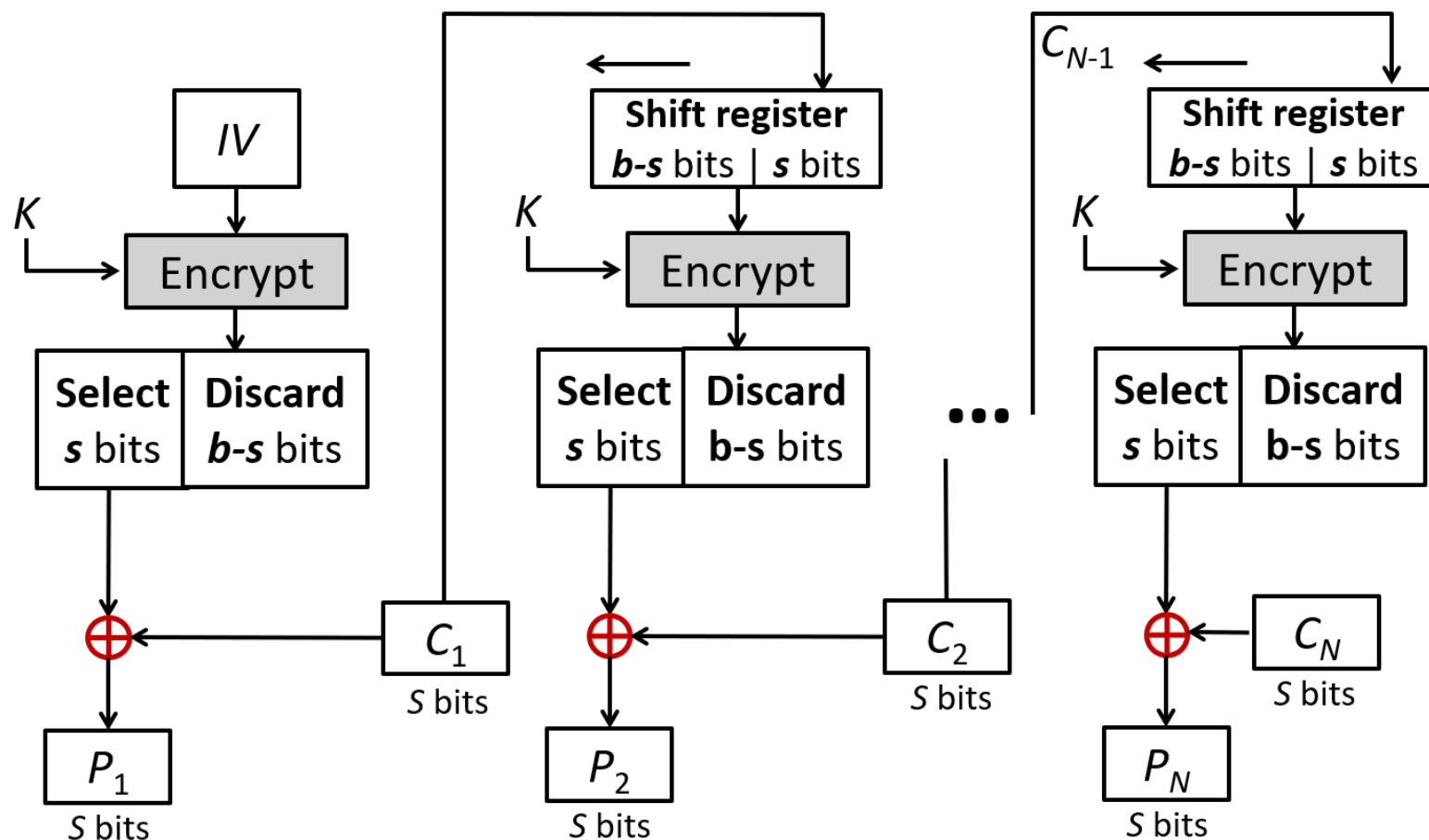
Cipher Feedback Mode:



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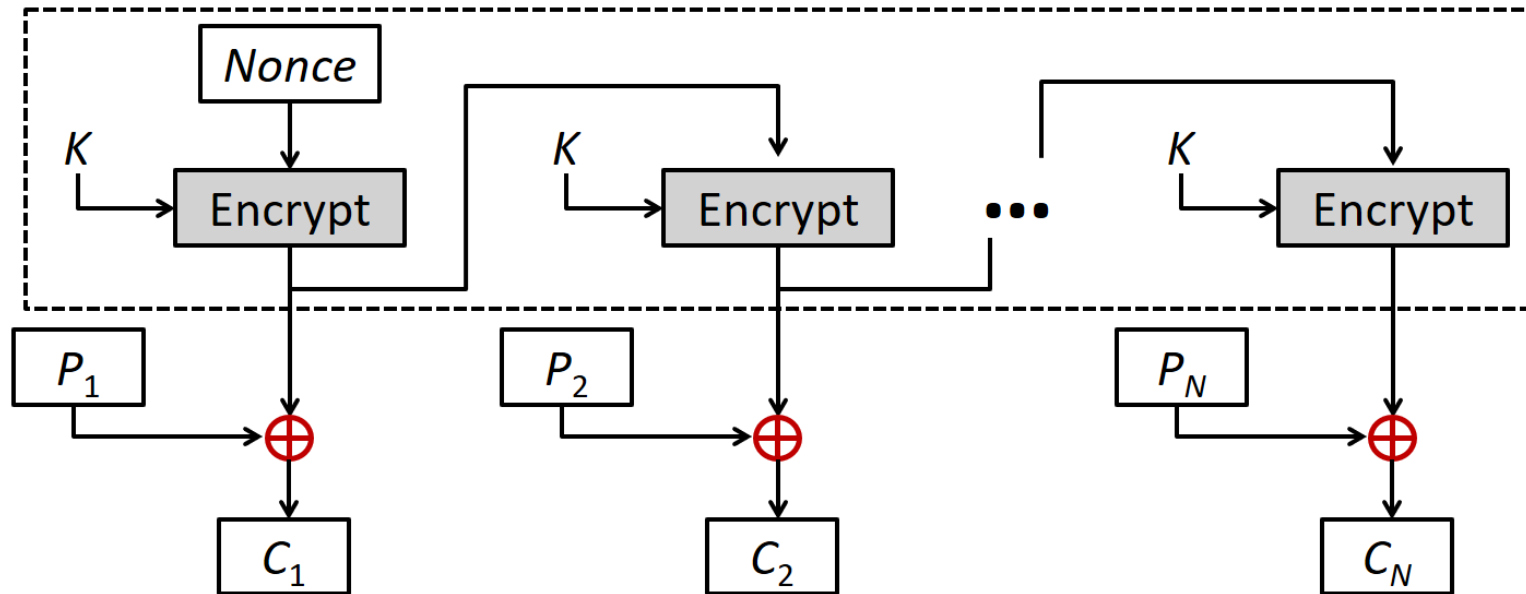
Cipher Feedback Mode:



Cipher Feedback Mode:

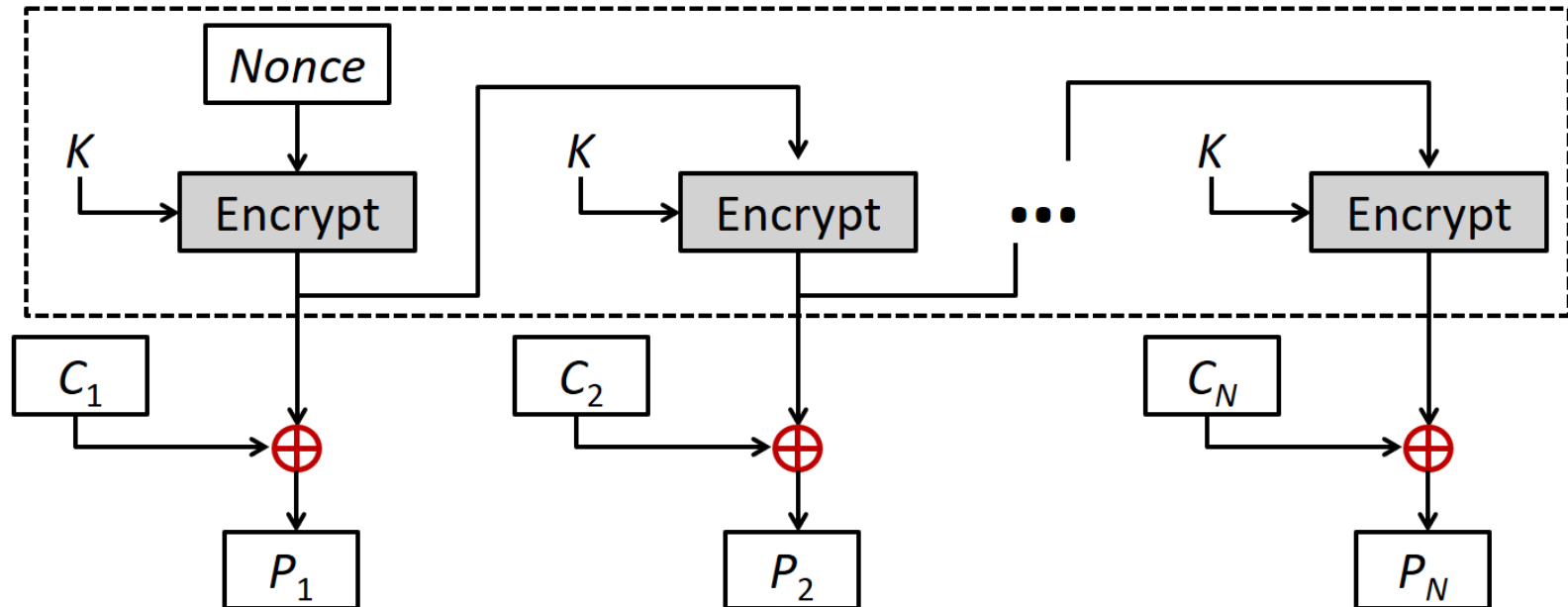
- The input to the encryption function is a b -bit shift register that is initially set to some initialization vector (IV).
- The leftmost (most significant) s bits of the output of the encryption function are XORed with the first segment of plaintext P_1 to produce the first unit of ciphertext C_1 , which is then transmitted.
- In addition, the contents of the shift register are shifted left by s bits, and C_1 is placed in the rightmost (least significant) s bits of the shift register.
- For decryption, the same scheme is used, except that the received ciphertext unit is XORed with the output of the encryption function to produce the plaintext unit.

Output Feedback mode - Encryption:



$$\begin{aligned}
 I_1 &= \text{Nonce} \\
 I_j &= O_{j-1} & j &= 2, \dots, N \\
 O_j &= E(K, I_j) & j &= 1, \dots, N \\
 C_j &= P_j \oplus O_j & j &= 1, \dots, N-1
 \end{aligned}$$

Output Feedback mode:



$$I_1 = \text{Nonce}$$

$$I_j = O_{j-1} \quad j = 2, \dots, N$$

$$O_j = E(K, I_j) \quad j = 1, \dots, N$$

$$P_j = C_j \oplus O_j \quad j = 1, \dots, N - 1$$

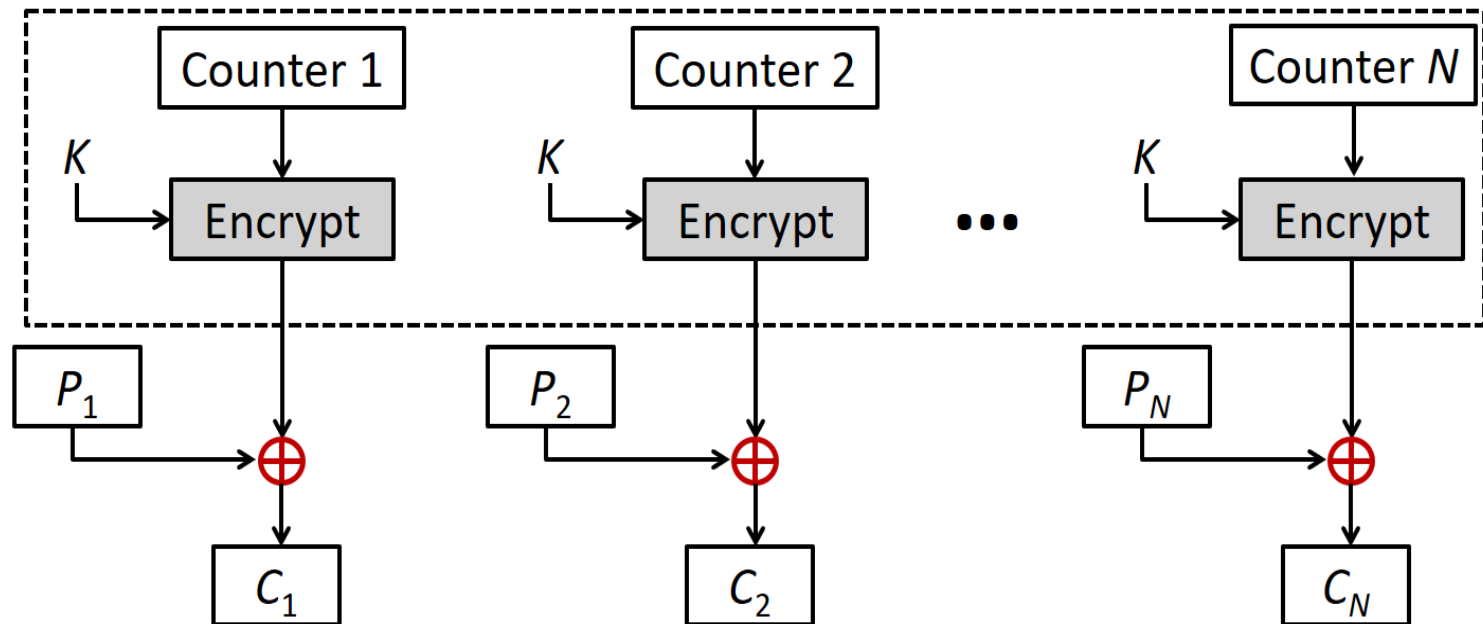
Output Feedback mode:

- The output feedback (OFB) mode is similar in structure to that of CFB.
- For OFB, the output of the encryption function is fed back to become the input for encrypting the next block of plaintext.
- In CFB, the output of the XOR unit is fed back to become input for encrypting the next block.
- The other difference is that the OFB mode operates on full blocks of plaintext and ciphertext, whereas CFB operates on an s-bit subset.

Output Feedback mode:

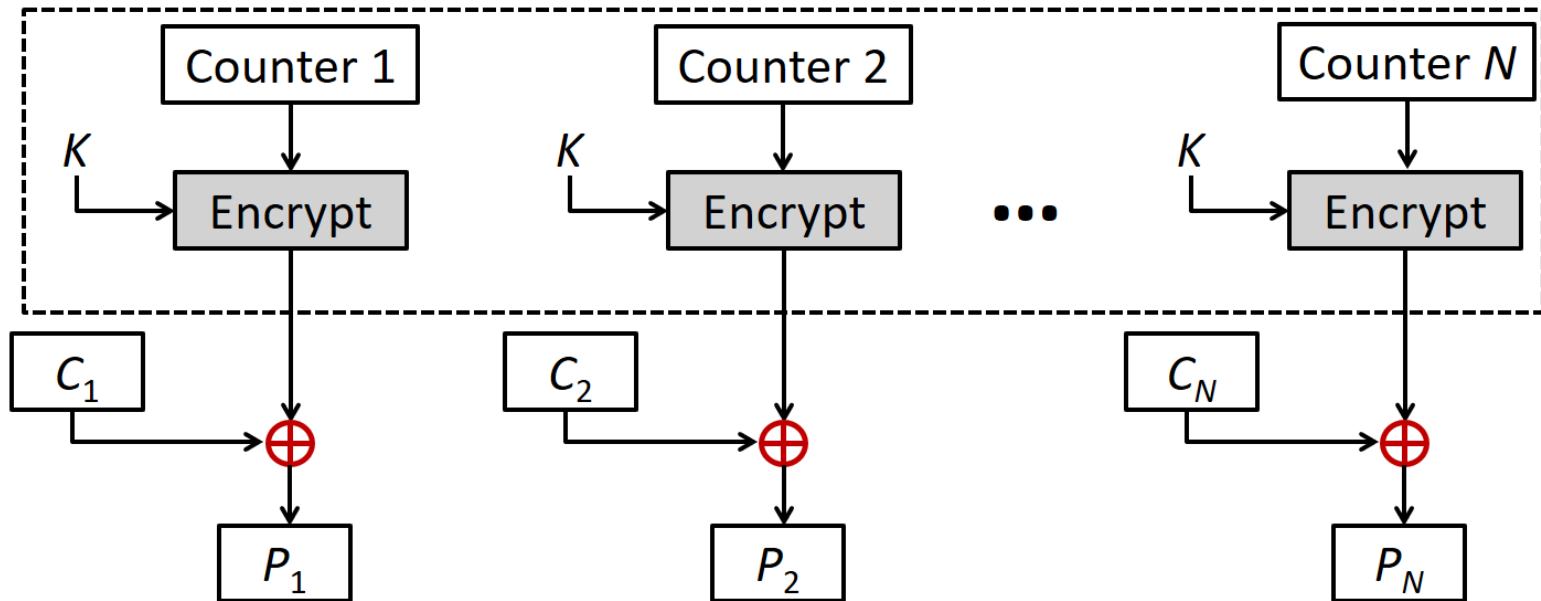
- **Nonce:** A time-varying value that has at most a negligible chance of repeating, for example, a random value that is generated anew for each use, a timestamp, a sequence number, or some combination of these.
- Each bit in the ciphertext is independent of the previous bit or bits.
- This avoids error propagation.
- Pre-compute of forward cipher is possible.

Counter Mode: Encryption



$$C_j = P_j \oplus E(K, T_j) \quad j = 1, \dots, N$$

Counter Mode: Decryption



$$P_j = C_j \oplus E(K, T_j) \quad j = 1, \dots, N$$

Counter Mode:

Counter (CTR) mode has increased recently with applications to ATM (asynchronous transfer mode) network security and IP sec (IP security).

A counter equal to the plaintext block size is used.

The counter value must be different for each plaintext block that is encrypted.

Typically, the counter is initialized to some value and then incremented by 1 for each subsequent block.

Counter Mode:

- **Strengths:**
- Needs only the encryption algorithm.
- Random access to encrypted data blocks.
- blocks can be processed (encrypted or decrypted) in parallel.
- Simple and fast encryption/decryption.

- Counter must be

Must be unknown and unpredictable.
pseudo-randomness in the key stream is a goal.

Counter Mode:

Operation Mode	Description	Type of Result
ECB	Each n-bit block is encrypted independently with same key.	Block Cipher
CBC	Same as ECB, but each block is XORed with previous cipher text.	Block Cipher
CFB	Each s-bit block is XORed with s-bit key which is part of previous cipher text.	Stream Cipher
OFB	Same as CFB, but input to the encryption is preceding encryption output.	Stream Cipher
CTR	Same as OFB, but a counter is used instead of nonce.	Stream Cipher

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